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REVIEW PAPER

Recent Advances on the Challenges of Resilient Automation Systems of the Power Grid: Critical Review

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Abstract

Despite their infrequency, natural disasters like hurricanes, tornadoes, and floods pose significant threats to power systems, with profound economic impacts on nations and individuals. This paper delves into enhancing power system resilience against such disruptions through techniques such as network reconfiguration, defensive isolation, and targeted strengthening. A critical factor in power system failures is cyber-attacks, which exploit the integration of sensors, monitoring tools, and communication devices. To bolster system defenses against cyber threats, strategies like intrusion prevention systems, multiple layers of defense, and prompt security responses are implemented. Additionally, ageing infrastructure and human error contribute significantly to system failures, underscoring the importance of error prevention and maintenance of aging components. This paper conducts a comprehensive analysis of the grid's advanced automation systems, reviews several case studies, and proposes solutions to mitigate the challenges posed by these threats.

Keywords Resilience · Power grid · Cyber threats · System failures · Prevention system

Introduction

Modern societies heavily rely on electric energy; its absence could lead to dire consequences. Power is essential for the prosperity of nearly every industry within a nation. However, the electricity system is consistently vulnerable to severe damages. These threats to power networks, which result in significant damages, arise from various sources including human attacks and natural disasters such as hurricanes, floods, and ice storms. The rarity and destructive magnitude of these natural disasters render them the most significant hazards to power systems. Consequently, research into power system resilience is becoming increasingly vital [1–11].

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Resilience refers to the ability of infrastructure to withstand and recover from disruptions or disturbances, as well as to proactively adapt measures to avert imminent threats. This definition varies according to the specific area of application. In the context of power grid resilience, while there is no standard definition, efforts by research institutions have led to some consensus. For instance, the United Kingdom Energy Research Center suggests that a resilient power system should have the capability to tolerate disturbances, swiftly recover, and continue supplying energy during external disruptions. Similarly, the American Society for Industrial Security defines power system resilience as the ability of system automation to resist disturbances and improve to an acceptable level of operation [1–10]. Resilience encompasses aspects of threat anticipation, resistance management, and recovery from disturbances. This review is driven by the critical role of the electrical grid in modern civilization, where it serves as the backbone for residential, commercial, and essential infrastructural operations. With the increasing reliance on automation, the grid has integrated advanced technologies for enhanced reliability and efficiency. However, this evolution brings new challenges, including vulnerabilities to cyber-attacks, natural disasters, equipment malfunctions, and human error, all of which compound the complexity of the electrical infrastructure. This study aims to provide valuable insights for scholars, policymakers, and industry professionals by critically evaluating current advancements in grid automation. It offers a comprehensive overview of the current state of the field, identifying areas requiring further research. Understanding technological advancements can guide policymakers in formulating effective strategies to support grid resilience, while industry professionals can gain practical insights to enhance grid stability and security through improved automation systems. Overall, this critical review serves as a timely and informative resource for stakeholders, enabling well-informed and adaptive decisions in the realm of power grid automation.

Literature Review

This section presents a comprehensive overview of a range of power resilience technologies and techniques, along with an analysis of the most relevant literature. To offer a robust understanding of power systems resilience, a resilience matrix, as depicted in Table 1, outlines key aspects such as planning, absorption, recovery, and adaptability in response to adverse events.

Traditional methods have been applied to achieve resilience in power systems, yet achieving absolute resilience remains a challenge [1, 11, 12]. Enhancing power system resilience involves adopting sophisticated methods and technologies such as the integration of distributed energy

Table 1 Scope of Resilience

Capability/ Domain	Plan	Absorb	Recover	Adapt
Physical	This considers the condition and the ability of equipment and operators, and its network mechanism	System awareness and response to maintain functionality	Normal functionality recovery	Resilience enhancement approach
Information	This deals with the preparation and presentation of data, analysis and storage	Anticipation and assessment of threats in real-time	Recovery data tracking and available set-ups	Design and expansion of information storage and use protocol
Cognitive	Robust plan considerations armed with decision-making and awareness abilities against threats	Proactive event management and contingency protocol	Decision-making towards recoverability and communication	New system design considerations or installation objectives and decision conditions
Social	Social network, capital, norms and training	Availability of capable personnel and adequate infrastructure	knowledge sharing and Teamwork to aid system recovery	Cultures, training programs, policies and institutional changes or adjustment

resources (DER), microgrids, and islanding. Resilience evaluation is divided into quantitative and qualitative methods, with the former involving simulation and statistical analysis [13, 14] (Fig. 1).

Significant advancements in the power industry have led to the adoption of sophisticated power grid automation systems to enhance grid resilience. These systems are predominantly cyberphysical, featuring advanced sensors, communication protocols, algorithms, interfaces, data processing capabilities, and networks. The integration of cyber-physical elements into power grids has increased vulnerability to human-perpetrated attacks. To address this, the U.S. Department of Energy has developed a novel approach for cyber-physical resilience, aiding in the analysis of past events and the formulation of control and response strategies, as illustrated in Fig. 2 [15–19].

Increased focus on power system resilience has led to collaborative efforts from research institutions and industry leaders to pre-emptively address and manage threats to power systems. This proactive approach stems from previous experiences of power system failures, including blackouts caused by both external events like natural disasters and internal component failures [13]. Examining the losses caused by these events underscores the critical importance of power system resilience. Hurricanes, as an example, cause extensive damage to systems, leading to catastrophic outcomes with considerable time and financial resources required for repair. Such damages not only inflict significant economic strain on countries but also adversely affect consumers [1, 20, 21]. Therefore, enhancing power system resilience is essential for both national and individual well-being.

This review conducts a comprehensive analysis of challenges and advancements in grid automation systems. It

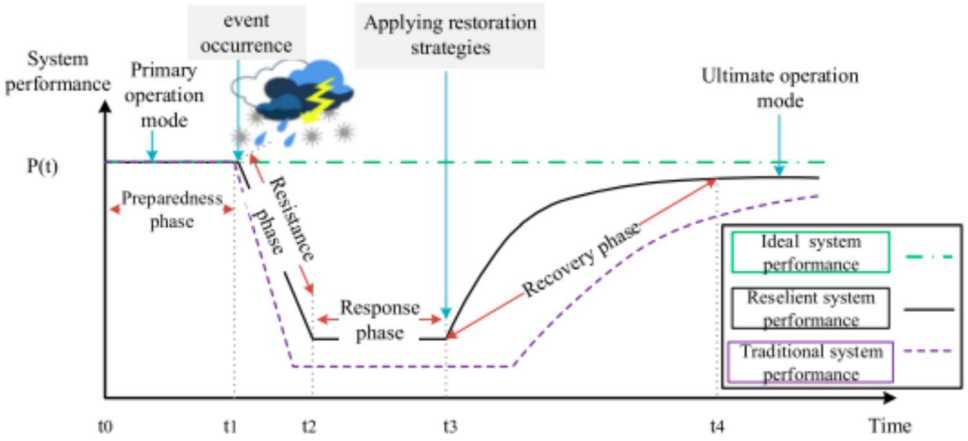


Fig. 1 Mechanism of Resilience under Disturbance [13]

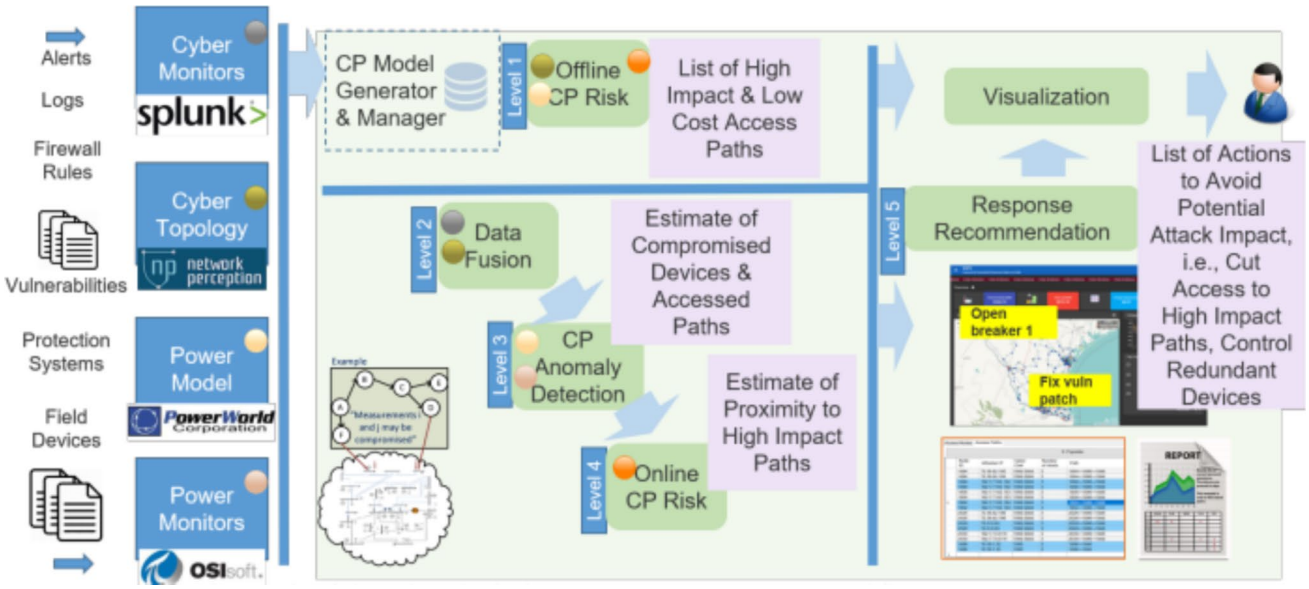


Fig. 2 Energy management flow for cyber-physical resilience of the grid [15]

explores a broad spectrum of challenges, including natural disasters, component failures, aging infrastructure, human errors, and intentional human attacks. Additionally, the paper investigates various approaches, methods, and technologies used to analyze, evaluate, and mitigate these threats. It delves into power grid automation innovations, highlighting advanced monitoring and sensing technologies, decentralized control mechanisms, artificial intelligence applications, and their importance in communication infrastructures, including 5G technology. This review also presents real-world case studies of resilient automation systems and examines existing regulations and policy considerations that pertain to the resilience of power grid automation systems. In conclusion, the paper identifies future directions and research opportunities in this area, offering valuable insights for further studies.

Challenges in Power Grid Resilience

Disruptions at even a single terminal of the network can precipitate a complete system breakdown, resulting in cascading failures and, in the worst-case scenario, a total blackout. Despite numerous efforts to bolster system resilience, challenges persist. Particularly in the case of natural disasters, the severity of their impact on power systems has escalated over time. This section outlines the most prevalent challenges confronting power systems, including natural disasters and extreme weather conditions, aging infrastructure, human error and unreliability, and cyber-attacks. It also discusses the latest methods and technologies employed to enhance power system resilience [22].

a) Natural Disaster and Extreme Weather Conditions

Natural disasters often arise from weather variations, though geological factors like earthquakes and volcanic eruptions, and human activities such as deforestation and urbanization, also play a role. A notable early instance was the New England Hurricane of 1938 that devastated Boston. Following this were several other significant events, including Hurricane Katrina (2005), Hurricane Sandy (2012), the 2011 Japan earthquake, and Hurricanes Hermine and Irma in 2016 and 2017, respectively. Hurricane Sandy, for example, disconnected about 7.5 million consumers across 15 U.S. states from the power grid. Similarly, Hurricane Irene in 2011 left 6.5 million people powerless. The 2008 ice storm in China caused estimated damages of $2.2 billion and left 200 million without electricity. These incidents inflict considerable damage to infrastructure and carry profound social and economic impacts, emphasizing the urgency of enhancing power system resilience. Other formidable natural threats include floods, short-lived but devastating tornadoes, and windstorms—particularly frequent in coastal regions—which severely impact transmission and distribution lines, as evidenced by significant economic losses in Guangdong Province, China. Ice storms, causing ice accumulation on power lines, increase line tension, potentially leading to breakages and outages. Some disasters, like Hurricane Maria, had effects lasting up to ten months [21, 23–33]. Tables 2, and 3 from [34] show a summary of the causes of blackouts, the outages and the number of people affected.

Table 2 Percentage of Blackouts around the Globe between 2011–2019

Blackout Cause	Sum of Number Recorded	Sum of % of the Recorded Number
Animals	1	1.50
Faulty equipment or human error	21	31.80
Over demand	4	6.10
Total	66	100.00
Vehicle/Accidents	7	10.60
Weather/Trees	33	50.00
Total	**132**	**200.00**

Table 3 Total Number of Blackouts in the USA between 2008–2015

Year	Sum of Total Number of Outages	Sum of People Affected (in millions)
2008	2169	25.80
2009	2840	13.50
2010	3149	17.50
2011	3071	41.80
2012	2808	25.00
2013	3236	14.00
2014	3634	14.20
2015	3571	13.20
Total	**24,478**	**165.00**

With the intensification of global warming and associated climate changes, the frequency of such natural disasters is likely to increase [35]. Numerous methods, approaches, and technologies have been developed to enhance the flexibility of power systems in responding to these events. These strategies aim to strengthen critical aspects of power systems for effective management and mitigation of future impacts [32]. Such methods include targeted hardening (such as replacing components with more robust alternatives) [36], defensive islanding [37], enhancing operational procedures and situational awareness [38], implementing redundancy and smart grid solutions, transmission line switching [32, 38, 39], adopting load flexibility-based methods [40], utilizing

network reconfiguration and decision-making restoration tools [25], deploying microgrids [24, 41, 42], and planning for energy storage [43].

b) Cybersecurity Threats

Cyber-attacks represent a form of malicious intrusion into the cyber-defense communication layers of power systems. Power systems play a crucial role in the economic progress of any society. In response to the increasing demand for sustainable, clean, and emission-free energy, and for efficient power system management and maintenance, numerous modifications and enhancements have been applied to traditional power systems. These include the integration of modern technologies, automation, control systems, state estimation, and energy management systems for monitoring, management, and optimization. Successful cyber-attacks can inflict severe damage to power systems and substantial losses to consumers. Distributed generation technologies like smart grids utilize these devices for wide-area surveillance, signal processing, and control of bulk energy grids. Such attacks may involve the injection of fake meter data, sensor compromise, and system state manipulation. Figure 3 illustrates a modern power grid's vulnerability to cyber-attacks [44–52]

The consequences of a successful cyber-attack can be far-reaching. In response, researchers have developed various frameworks and techniques to enhance the resilience of power automation systems against such cyber threats. These include the National Institute of Standards and Technology's (NIST) cybersecurity framework, as shown in Fig. 4, and other strategies such as a layered framework for smart grid automation system protection against cyber-attacks [62], a four-layer attack-defense framework focusing on security enhancement during the design process [63], the integration of anomaly detection devices and attack mitigation systems [55], time-based security mechanisms [58], and intrusion prevention systems [64].



Fig. 4 The NIST's cybersecurity framework [53]

c) Human Errors and System Failures

The human element plays a crucial role in any power system. From equipment manufacturing and dispatch to system operation and maintenance, human involvement is pervasive. Consequently, many power system failures are attributable to human error. Notable examples include the US-Canada

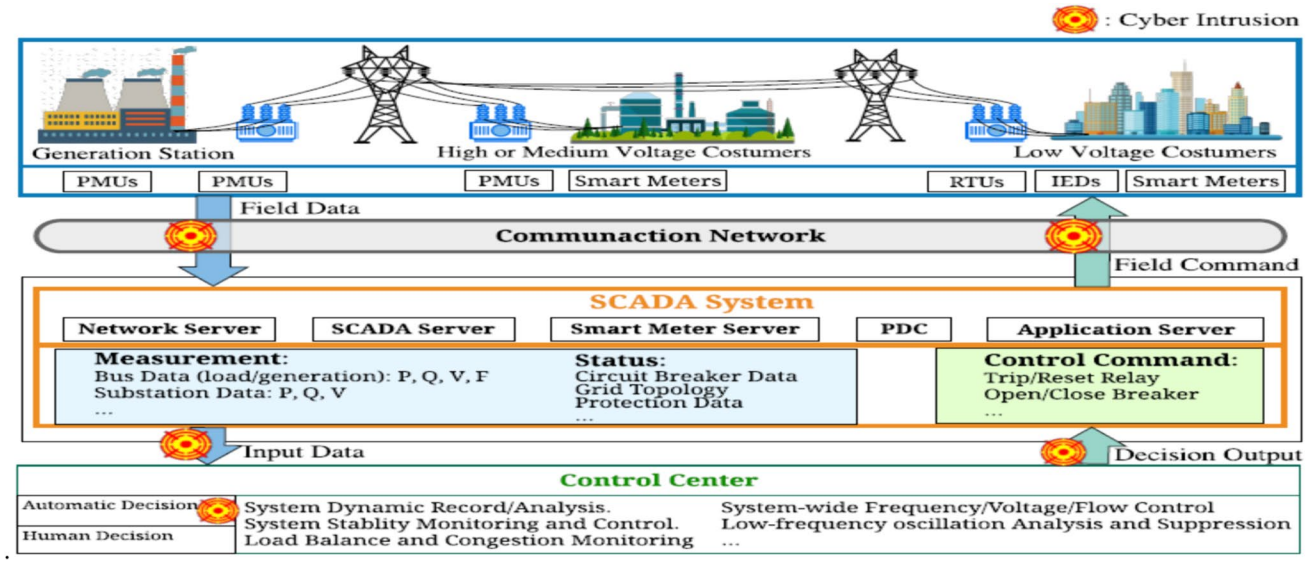


Fig. 3 Cyber Vulnerabilities of a Modern Power Grid [45]

blackout of 2004, the California blackout of 2011, and the Italian blackout of 2003. Human error refers to any action or inaction that leads to failure or damage within the power grid/system. The introduction of new technologies and advanced devices has significantly increased the complexity of information, communication, and control systems. This complexity raises questions about the cognitive abilities and qualifications of power system operators. Can they efficiently understand and operate these advanced systems, and make quick, effective decisions in critical situations? It is, therefore, reasonable to assert that human error is an inescapable aspect of power system operation. These errors can be classified as omissions, unnecessary actions, sequence errors, design flaws, fabrication mistakes, maintenance oversights, and inspection errors.

Figure 5 illustrates the relationship between human error, task complexity, and operator experience [54]. Factors such as psychological stress, task complexity, fatigue, approach and sense of obligation, incorrect actions, inaccuracy, delayed execution, negligence, lack of knowledge and experience, poor concentration, inadequate training, ineffective production management, absence of a safety culture, poor communication skills, lack of teamwork, unfavorable working conditions, decision errors, misdiagnosis, delayed interpretation, carelessness, illusion, and nervousness can contribute to human error. Human factors are also significant causes of component failure, including improper arrangement, incorrect maintenance procedures, operating beyond design specifications, inadequate monitoring, and failure in design analysis [54–64].

A robust framework can be employed to minimize human error in power systems [57]. This framework focuses on managing human attention by incorporating dynamics like the special layout, optimized brightness and contrast in the working environment, and the use of resilient code. It also emphasizes the importance of feedback mechanisms to better inform operators, methods to lock out erroneous actions and incorrect inputs, and providing ample support and assistance to operators.

d) Aging Infrastructure

The power grid automation system consists of a complex network of interconnected equipment and devices. Over time, these components undergo wear and tear, increasing their likelihood of failure. Such failures significantly impact the reliability of the power system, leading to reduced overall performance, more frequent downtime, and decreased efficiency. For example, aging can diminish a component's thermal capacity, impairing its functionality, increasing vulnerability, and raising maintenance costs. These are common consequences of aging infrastructure in power grid automation systems, often resulting in accelerated degradation. Aging is a primary cause of component failure, as depicted in Fig. 6 [65–70].

The adverse effects of aging components on power grid automation systems can be mitigated to enhance resilience. One approach is adopting the Time of Use model [71] to adjust load schedules based on the thermal condition of components. Strategies such as proper maintenance and targeted component replacement are proposed [68, 69] to improve system performance and overall resilience. An online impedance detection method for equipment aging [72] can be used to enhance the reliability of transmission lines. Effects of these disasters on power systems carry significant economic costs for countries and individuals. Measures like targeted hardening, defensive islanding, and network reconfiguration are methods to bolster resilience against such events. Cyber threats, classified as intentional attacks, significantly contribute to power system failures. Defense strategies, including intrusion prevention systems, multi-layered attack defense, and time security mechanisms, are vital for enhancing resilience against cyber threats. Other factors leading to power system failure encompass human errors and infrastructure aging.

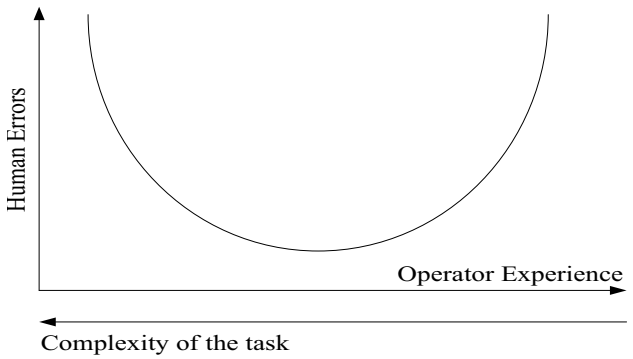


Fig. 5 Relationship between human error, complexity of task and operator experience [54]

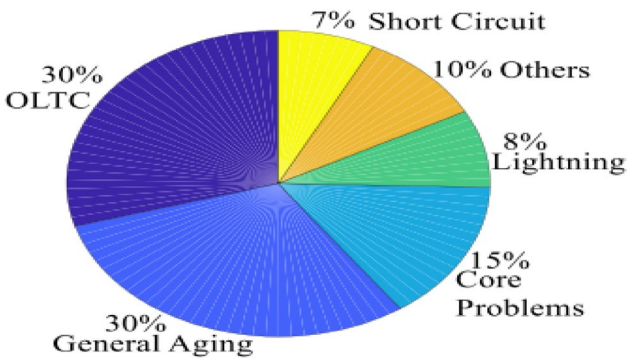


Fig. 6 Causes of component failure [65]

Innovation in Resilient Automation Systems

a) Implementation of IoT and Smart Sensors for Real-Time Data Collection

Internet of Things (IoT) has revolutionized device usage, enabling advanced control and management of appliances, vehicles, smart homes, and various equipment through real-time data collection. This capability largely depends on smart sensors that gather data points such as temperature, humidity, vibration, and operational conditions. These sensors transmit the data via the internet using specific communication protocols to either a cloud or a local server. The latter is equipped with computational and analytical capabilities to process and analyze the data, identifying patterns or clusters. This processed data can then initiate actions or await commands from a user interface, which might be a mobile or desktop application. In smart environments, like homes or ecosystems, extensive communication occurs among numerous devices and interfaces. They interact with each other and possibly a central unit or server, coordinating control and management efforts. Figure 7 illustrates the various layers and processes involved in IoT [73–81].

Advancement in power system automation is integrating IoT technology, leading to the development of smart grids. This integration allows for instantaneous data collection from various parts of the smart grid for advanced metering and monitoring, a capability not feasible in traditional power grids [76, 82]. This is further enhanced with big data technology [75].

Research by [83] focused on reducing machine downtime due to failures, employing IoT for failure prediction and monitoring power consumption. They introduced power monitoring system that is dependent on IoT to detect abnormal power usage in machines, offering remote data access via Android apps and cloud storage. However, this system's reliance on internet connectivity is a notable limitation. In contrast, [84] advocates for the use of edge computing to process and analyze data near its source, reducing reliance on centralized cloud infrastructure and mitigating latency issues in remote data transmission.

Reference [85] identifies unexpected failures at the transformer level as a key cause of power system blackouts. They recommend installing IoT devices at transformer levels, as transformers are critical network feed units. This approach, they argue, achieves operational efficiency and cost-effectiveness. The authors developed a cost-effective IoT monitoring system tailored for power grid transformers, enabling real-time tracking of their operational status, including temperature, humidity, oil level, voltage, vibration, and pressure.

In study [86], IoT technology was deployed for remote load management within a smart grid, utilizing a smart energy meter to enable users to monitor and regulate energy usage in their homes. This work, however, solely concentrates on remote load management in smart grids, extending control to household appliances and events aimed at reducing energy costs. A significant limitation is its inability to monitor non-electric devices. In addition, the smart energy meter provides users with instant power consumption data, facilitating real-time load management through IoT capabilities. A mobile application offers a visualization of this

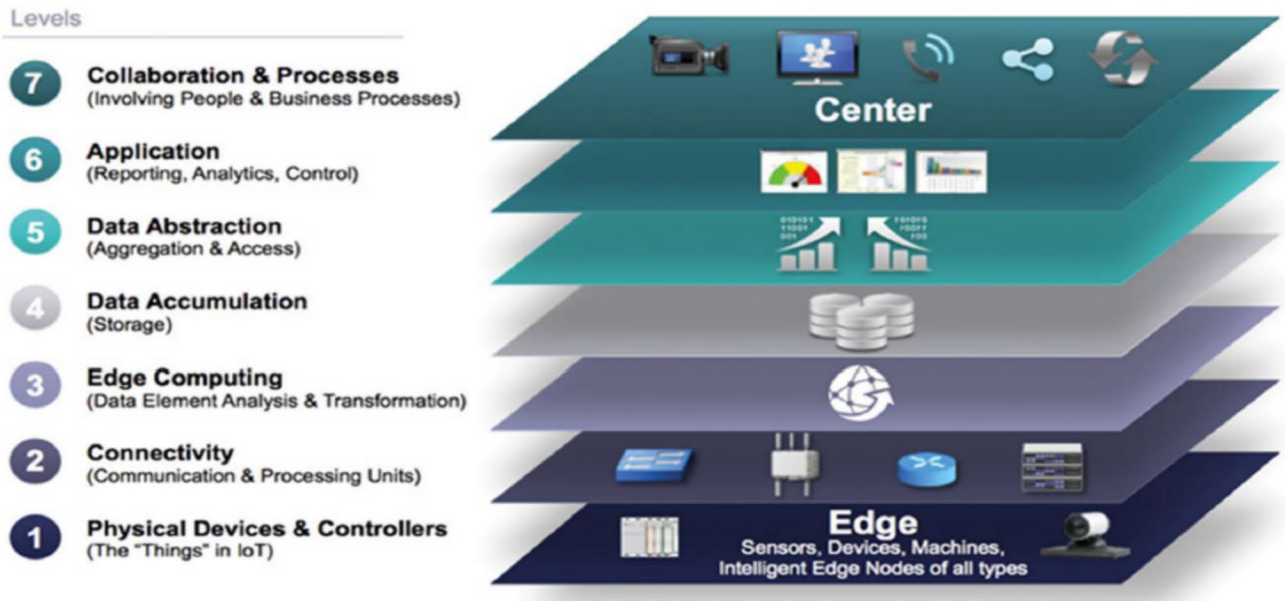


Fig. 7 The IoT World Forum reference model [82]

Table 4 Advantages of smart grid over traditional power system with the help of IoT [88]

	Traditional Grid	Smart Grid
Data flow	One-way communication	Two-way communication
Electricity generation	Centralized generation	Distributed generation
Grid topology	Hierarchical	Network
Sensors	Few sensors	Lot of sensors
Monitoring	Blind	Self-monitoring
Outage recovery	Manual restoration	Self-healing
Testing	Manual check/test	Remote check/test
Type of Control	Limited Control	Pervasive control
Data flow	One-way communication	Two-way communication
Electricity generation	Centralized generation	Distributed generation

real-time energy monitoring. Despite these advantages, a notable drawback is the privacy and security concerns, which can be addressed by implementing data encryption, access controls, and anonymization techniques. According to [88], IoT represents a significant advancement over traditional SCADA systems by incorporating wireless technology for real-time data acquisition. The researchers developed a system using an Arduino connected to various sensors to monitor and address issues in the transmission system, simulating circuits with multiple sensors to observe and manage potential technical issues.

Moreover, combining IoT with power line carrier communication technology offers several benefits to smart grids compared to traditional grids, as detailed in Table 4 [88]. IoT is utilized to monitor transmission line operations in real-time (Fig. 8 [89, 90]), enhance power quality [74], improve security [77, 91], and facilitate cloud-based data acquisition [80]. Additionally, IoT has been implemented for online monitoring of a portable off-grid solar system [78], as illustrated in Fig. 9.

b) Utilization of Big Data Analytics for Power System Resilience

The shift from traditional grids to micro-grids and smart grids involves installing numerous modern devices that measure and transmit large volumes of information [92]. Diverse data formats and varying speeds of collection and transmission present a challenge in managing, processing, and analyzing these datasets. This complexity can result in delayed responses and inadequate equipment maintenance, increasing downtime [93].

According to [94], 'Big Data' refers to the use of a data collection infrastructure with substantial computational capacity to access data that are typically challenging to retrieve from conventional databases. This research explores various Big Data applications within power systems, emphasizing how Big Data analytical technology enhances power and energy systems development. This enhancement is achieved through integrating and managing electric power within extensive data networks, employing sophisticated data processing technologies and visualization for in-depth analysis. Moreover, integrating Big Data into power systems allows for the use of multiple data sources, yielding more reliable, accurate, and actionable information, significantly surpassing traditional methods [95].

Furthermore [94], notes that Big Data can handle varied data formats transmitted at different speeds (refer to Fig. 10). Efforts to reduce power grid automation system downtime include predictive maintenance, effectively implemented through Big Data applications. This method surpasses

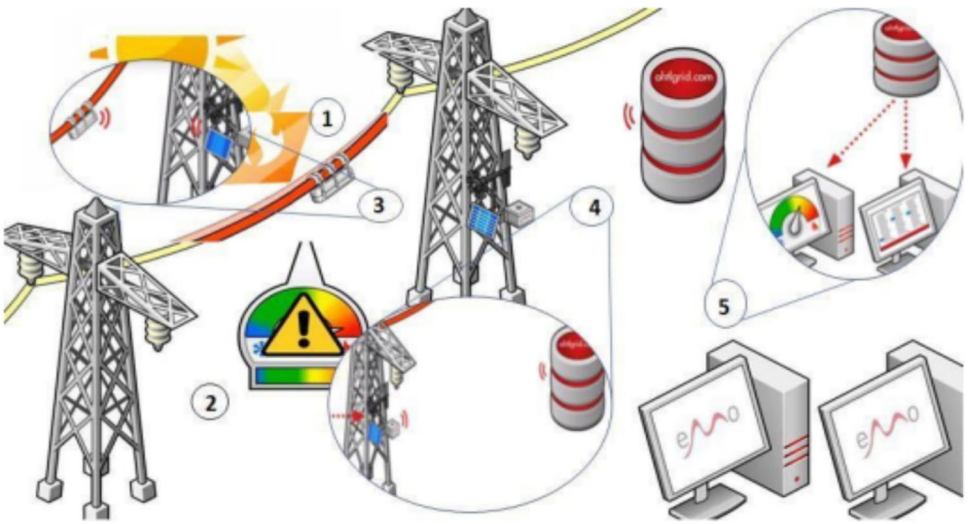


Fig. 8 Real-time monitoring transmission line operation using IoT [89]

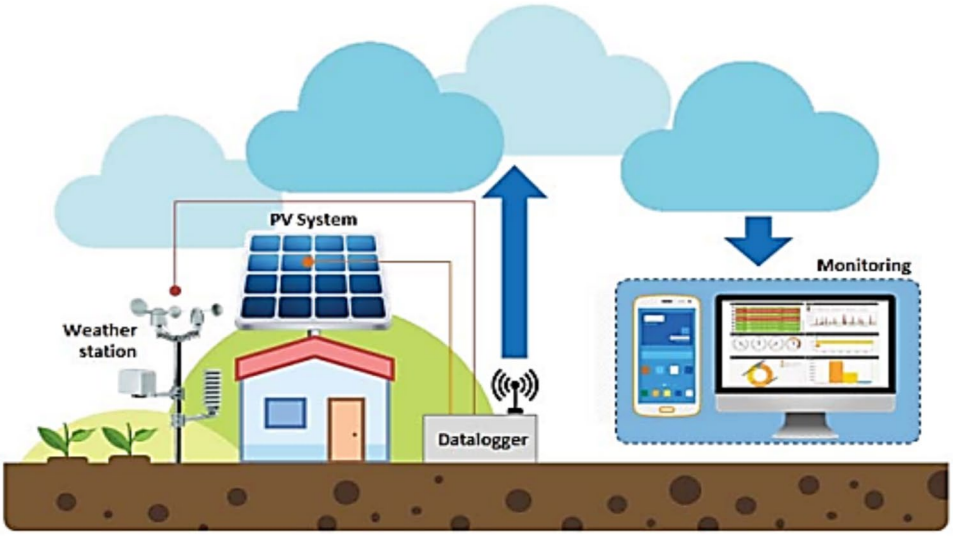


Fig. 9 Online (web) monitoring of an off-grid solar system [78]

reactive or scheduled maintenance in efficiency and cost-effectiveness [96]. Big Data's predictive maintenance not only offers economic benefits to power retailers (as depicted in Fig. 11) but also finds wide-ranging applications in power systems, as demonstrated in Fig. 12 [94, 97, 98].

c) Decentralized and Distributed Control

The increasing load on power systems presents a substantial challenge. Centralized grids struggle to effectively control, monitor, and make decisions across extensive networks. This situation calls for the decentralization of energy generation and management, including distributed micro and nano grids. Decentralization enhances power infrastructure management and control, offering multiple advantages: reduced reliance on central power control centers, thereby increasing grid resilience; localized fault detection; rapid response to faults; prevention of cascading failures during natural disasters; minimized ageing due to reduced central control unit stress; efficient regulation and load balancing; enhanced scalability; diminished vulnerability to cyber-attacks; customer autonomy in energy consumption control; and reduced transmission losses [99–107].

Reference [108] advocated for connecting two power-generating units, such as Solar Photovoltaic (PV) and wind turbines, with an advanced frequency control system for grid decentralization. In this model, each unit functions either as a primary source or a backup. This setup, however, is prone to climate and weather-induced power fluctuations, potentially leading to inadequate load coverage. Integrating a battery energy storage system (BESS) can counter these vulnerabilities, fortifying the system against climate-related challenges.

To enhance Demand Response (DR) and load control, [109] recommended the adoption of Blockchain technology and peer-to-peer energy trading. This method allows direct electricity trade between consumers via Blockchain's peer-to-peer feature, enabling real-time transaction adaptation to current demand and supply. Such a system empowers consumers to capitalize on their energy flexibility, simultaneously bolstering the grid's resilience and sustainability. Nonetheless, the implementation faces challenges like digital literacy and potential cyber offenses.

Automation systems in power grids are equipped for autonomous recovery, optimization, and protection from

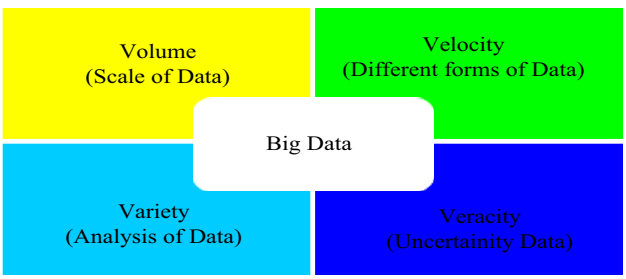


Fig. 10 Features of Big Data [94]

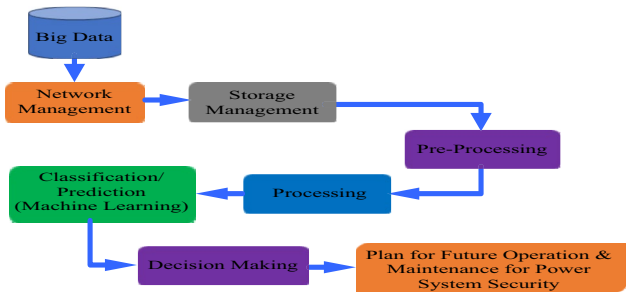


Fig. 11 Big Data for predictive maintenance [94]

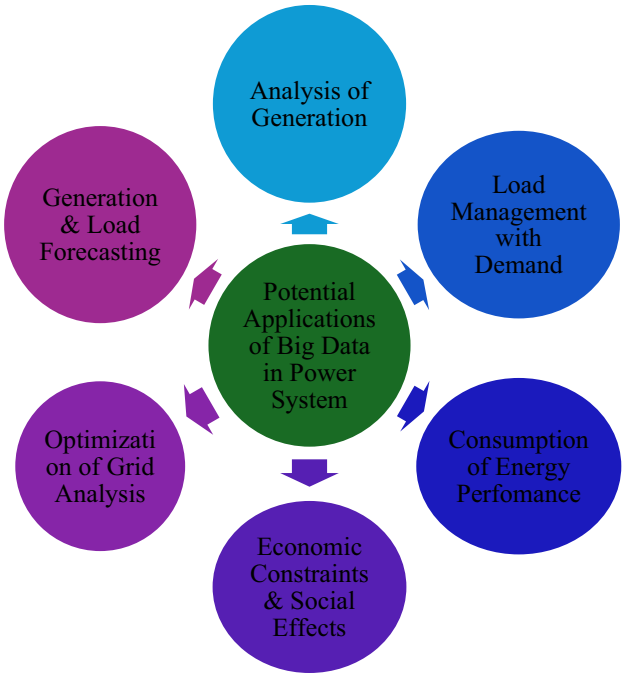


Fig. 12 Full scale of Big Data application in power grid [94]

physical and cyber threats. They can redirect power flow to sustain service in fault situations. Decentralizing control leads to localized advanced automation systems, facilitating efficient self-healing. These systems ensure balanced load distribution, preventing resource overuse or underuse in low-demand periods [110, 111]. Renewable energy sources further support self-healing [112]. Additionally, [113] introduces a decentralized, reactive multi-agent system algorithm for smart grids, integrating multistage processes [114].

d) AI / Machine Learning in Grid Automation

This section encompasses machine systems that learn, leverage knowledge and experiences, and engage in creative and inventive tasks akin to human thinking [115]. It involves solving complex problems and applying abstract concepts without needing exhaustive options or strategies [116]. These technologies' capabilities are applied in power systems for multiple purposes: forecasting renewable energy generation control, ensuring effective load balancing, accurately predicting electricity prices, enhancing customer participation, fostering enhanced decentralized demand responses, and detecting anomalies and preventing faults [117, 118]. An example of a model demonstrating improved security against cyber-attacks is shown in Fig. 8. Additionally, AI is instrumental in power grid automation for managing distributed energy sources, optimizing reactive power, predictive maintenance, making real-time decisions, improving situational awareness, and providing analytics and diagnostics. This enhances overall system control, state estimation, islanding detection for effective disturbance detection and cyber threat mitigation, as depicted in Fig. 13 [119–126].

Reference [127] discusses how the improvement in power grid technologies and integration of smart and modern devices may present challenges for power networks, including disturbances from natural and human-made events. The authors highlight the critical need to accurately identify disturbance types, as this impacts prediction and response strategies. To address this, they propose a machine learning-based attack detection model for power systems, trained using data and logs from phasor measurement units (PMUs). The model involves constructing features from the data, inputting them into various machine learning models, with

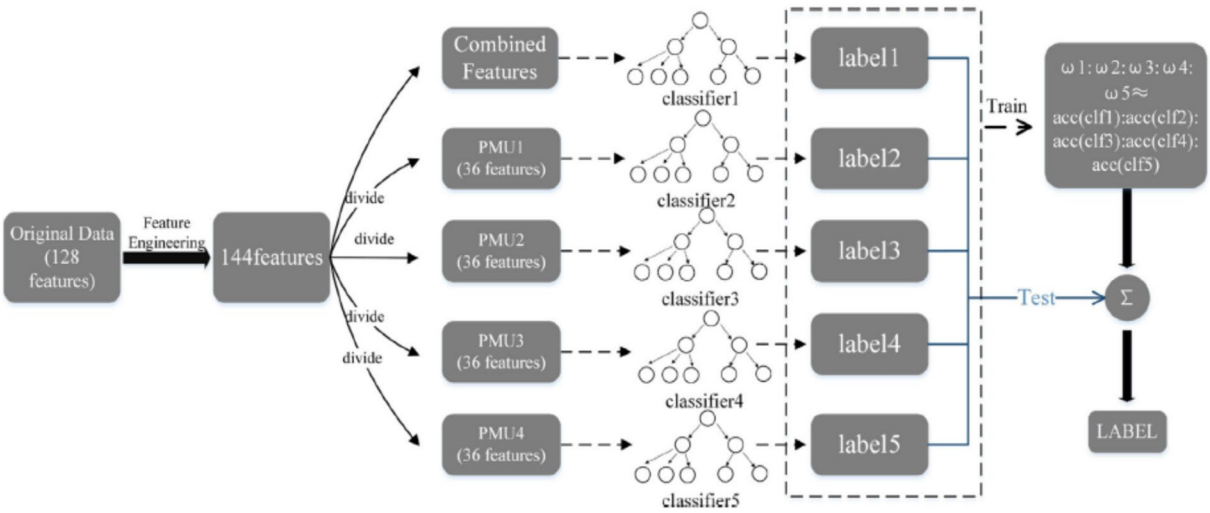


Fig. 13 Artificial Intelligence and Machine Learning in Detection of Disturbance [127]

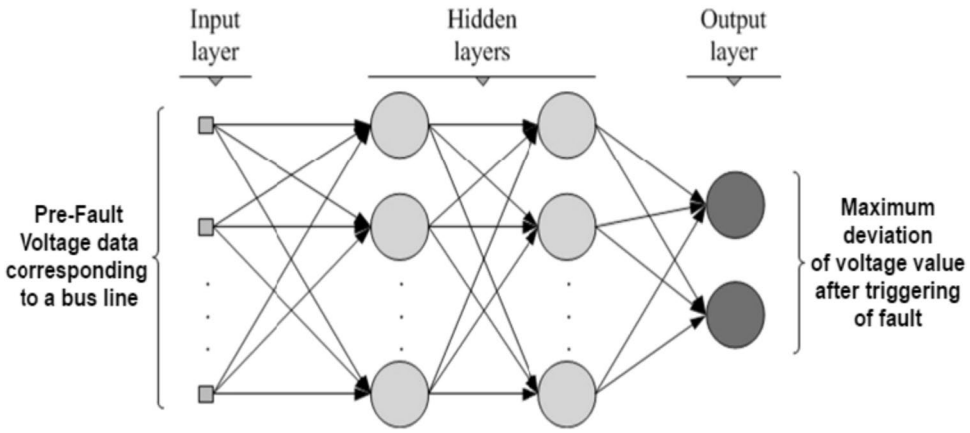


Fig. 14 A model for fault detection [128]

random forest (RF) as the primary classifier within AdaBoost for categorization. Its performance is assessed using simulated data from open-source platforms, illustrated in Fig. 13.

Further, [127] underscores the significant risks associated with even minor malfunctions in power grids, including potential productivity and revenue loss, and in some cases, loss of life. They stress the importance of developing a system for proactive grid health monitoring. To meet this need, an AI model using a recurrent neural network (RNN) assesses the grid's condition, as depicted in Fig. 14.

e) Resilient Communication Networks

The essential role of communication infrastructure in strengthening power grid resilience cannot be overstated. Effective power control relies on the communication system's capacity to transmit control and monitoring messages. A reliable communication infrastructure guarantees operational continuity under abnormal conditions, enabling effective monitoring, control, and self-healing. It is also instrumental in preventing significant damage to crucial power grid installations by efficiently detecting anomalies and responding promptly. The advent of low-latency 5G technology facilitates rapid and efficient communication among various smart installations, such as smart sensors, phasor measurement units, and IoT devices, thereby enhancing grid resilience. This technology offers extensive communication coverage, crucial for effective supply control and management in both urban and remote areas. Moreover, a robust communication infrastructure fortifies the grid against cyber-attacks through secure message authentication, system analysis, data integrity, and state monitoring, while also improving scalability and situational awareness [129–137].

According to [138], the integration of terrestrial and satellite links into a hybrid communication network can greatly boost power system resilience. This approach offers redundancy, wider coverage, efficient resource utilization, and enhanced monitoring capabilities. The study delves into the implications of this framework for power system resilience, introducing an adaptive link selection protocol at the network layer to enhance network integrity. Additionally, a modified transport protocol, tailored for satellite communication, has been proposed to optimize limited bandwidth usage. However, the implementation of this comprehensive approach may be hindered by associated costs and complexities.

f) Automated Distribution Management Systems (ADMS)

Automated distribution management (ADM) systems integrate technologies within power grids to automate tasks such as real-time monitoring, control, and management of distribution networks. These systems aim to boost the efficiency and resilience of power grids by continuously monitoring network conditions in real time, using advanced fault detection algorithms to identify faults and assist in the rapid restoration of grid operations [139–141].

The incorporation of distributed generation (DG) into contemporary power linkages introduces further tasks for protection systems. This is particularly true with bidirectional power flow that affects the settings of overcurrent relays (OCRs), as noted in [142]. This bidirectional flow can alter the sensitivity and response time of OCRs, potentially leading to malfunctions. To combat this, an algorithm for fault detection, islanding detection, and power swing management in radial DG systems was proposed. This algorithm capitalizes on phasor measurements, benefiting from the increased use of phasor measurement units (PMUs) in modern electrical grids. It employs positive sequence measurement techniques for fault detection and computes three crucial indices for this purpose. Capable of identifying and processing high severity faults, this algorithm issues signals to prevent malfunctions and activate emergency responses. The

algorithm's effectiveness, however, could be diminished if a PMU fails. To bolster this system, [143] enhances the original algorithm by integrating it into the Fault Location, Isolation, and Service Restoration (FLISR) program, which utilizes data from various wireless sensors and smart meters. This multi-faceted approach, characterized by its redundancy, ensures reliability and prevents dependency on the output of any single device or unit.

g) Energy Management System (EMS)

The energy management system integrates a suite of software, hardware, and communication infrastructure utilized by utilities or power system operators. Its fundamental role is to monitor, control, and optimize the functioning of the entire power system network, which encompasses generation, transmission, and distribution components. This system empowers utilities and power grid operators to implement automation algorithms, make informed real-time decisions, and exert control, thereby enhancing the resilience of the power system [144–146].

Incorporating renewable energy sources (RESs) into the power grid is a critical strategy for achieving efficiency, sustainability, and optimization in energy management [147]. Research highlighted by [148] addresses operational challenges arising from the rapid increase in uncontrollable RESs. To tackle these challenges, the study proposes an innovative dynamic load dispatch method focused on scheduling the outputs of generating units. This method is designed to mitigate irregularities linked with RESs and effectively manage real-time load demands. The proposed algorithm aids in unit commitment and dynamic economic load dispatch, optimizing both day-ahead and real-time generation schedules, even amid inherent power system uncertainties. However, the study does not account for potential deviations in the predicted source load of committed units and their effects on grid resilience and reliability. Addressing this, [149] presents a multiple-time-scale (MTS) rolling optimal dispatching framework aimed at evaluating and balancing the confidence probabilities of source-load power for robust, real-time scheduling plans. A plethora of modern technologies has been adopted in power systems to meet the increasing demand for power, clean energy supply, customer engagement in energy management, and overall system resilience. The Internet of Things (IoT), Big Data, decentralized and distributed control systems (such as smart grids), artificial intelligence, 5G, and satellite technology are prime examples of this modern infrastructure. These innovations have significantly bolstered the resilience of power systems, enhancing their capabilities in efficient sensing, monitoring, metering, and energy management (Table 5).

Case Studies

This section examines various implementations of power grid automation systems that have bolstered power system resilience.

a) Gulf Petrochemical Industries Company

In Bahrain, Gulf Petrochemical Industries Company specializes in producing methanol, ammonia, and urea. A failure in the ammonia plant section could lead to a total plant shutdown, incurring substantial costs. To mitigate this, the company has implemented an automatic decoupling system (ADS), equipped with a Sequential Events Recorder (SER) and enhanced monitoring capabilities. This system automatically isolates critical plant systems during external disturbances. Additionally, the company addresses vulnerabilities in the combustion gas turbine caused by national grid disruptions by using separate utility and combustion gas turbine feeds for redundancy. The installation of the state-of-the-art ADS ensures effective management of disturbances, particularly those intensified by the utility power infrastructure's expansion. The ADS, with its advanced diagnostic and analytical features, promptly isolates the combustion gas turbine from the utility during external disturbances. It monitors current and voltage levels, analyzes data, and triggers necessary disconnections to safeguard the Gulf Petrochemical Industries Company's network. This operation, assisted by the observed measurements at the interface location, ensures the synchronization of protective mechanisms and automatic event recording, covering both analogue and digital signals. The system's real-time monitoring capabilities, supported by a Sequential Event Recorder and remote SCADA control, have proven effective in isolating the company's systems during multiple incidents. Notably, the ADS have operated flawlessly, with no recorded equipment failures [150].

b) Southern Califonia Edison

Southern California Edison utilizes Remedial Action Schemes (RAS) to ensure reliable power performance following disruptions in the transmission grid network. These RAS systems serve as cost-effective alternatives to more expensive measures, such as reconductoring transmission lines, constructing new lines, or adding additional transformers. The RAS systems can detect, alert, compute, and resolve issues within a swift 50-ms window, substantially enhancing grid reliability. This is achieved by comprehensively monitoring grid parameters and responding quickly to anomalies. Designed for use across Southern California Edison’s operational scope and in conjunction with neighboring utilities, these systems significantly improve load management. Since implementing the Remedial Action Schemes,

Table 5 Summary of Reviewed Related Works

Related Work	Weakness	Proposed Solution
[52]	Concentrates only on cyber-physical security	Should include a robust resilient strategy
[54]	Specific to Electrical transmission network only	Should be expanded to generalize other types of power system (Generation and Distribution)
[55]	Limited to only nuclear energy power plants and digital control rooms	The integration of other sources of energy generation (renewable and hydro plants) and the inclusion of other control rooms (manually-controlled rooms)
[57]	Focuses on nuclear power plant operators only	Should incorporate different power plant operators e.g., hydro, gas plant
[68]	Focuses only on cost-effectiveness of the equipment	Other aspects like environmental impact assessment should be considered
[70]	Focuses on the impact of electric vehicle charging	Should consider other devices that will are connected to the power distribution transformer
[83]	Reliance on internet connectivity	Leveraging edge computing to decrease reliance on centralized cloud infrastructure
[85]	Deployment of IoT devices on every machine	Installation of IoT devices at the transformer level for efficiency and cost-effectiveness
[86]	Privacy and security concerns,	Employing data encryption, access controls, and anonymization techniques
[87]	Limited to Electricity Consumption Monitoring	Expansion of the existing architecture to monitor other parameters
[88]	Limited applicability of traditional SCADA systems	Integration of wireless technology and real-time data acquisition through IoT
[94]	Inadequacy of conventional database systems	Utilization of Big Data analytical technology for enhanced data integration and analysis
[97]	Limited to specific use of big data for prediction purposes	Other data sources outside the big Data should be used for training and prediction
[98]	Focuses on micro grid only	Should consider other power system network like nano-grid and national grid
[99]	Concentrates only on DC nano-grid and developing region	Incorporation of bigger grids like national grid as well as micro grid
[101]	Focus only on distribution networks	Should consider broader power system network
[108]	Susceptibility to climate and weather conditions	Integration of battery energy storage system (BESS) to bolster resilience
[109]	Concerns regarding digital literacy and cyber-related offences	Application of blockchain technology and peer-to-peer energy trading with enhanced security measures
[124]	Specific to information systems operations to power systems	Large data should be applied during the training process to enhance the efficiency of the results
[127]	Motivated by the need for power system resilience	Development of an artificial intelligence model for proactive grid health monitoring
[142]	Impact of bidirectional power flow on protection systems	Proposal of fault detection algorithm for distributed generation systems
[148]	Lack of consideration for deviations in predicted source load	Introduction of a multiple-time-scale (MTS) rolling optimal dispatching framework for robust scheduling
[149]	Motivated by the need fo effective scheduling of generating units	Assessment and balance of confidence probabilities of source-load power for robust scheduling

Southern California Edison has achieved a notable reduction in system downtime [151].

c) The Power System in Guatemala

In Guatemala, a wide-area supplementary control scheme (SCS), based on Phasor Measurement and Control Units (PMCUs), is employed for the power system's operational stability. This scheme primarily functions to separate the Guatemalan power system from its southern region under unstable operational conditions. The strategy incorporates time-synchronized measurements of active power and modal analysis of active power flow between Guatemala and El Salvador. Through detailed analysis of the power system's

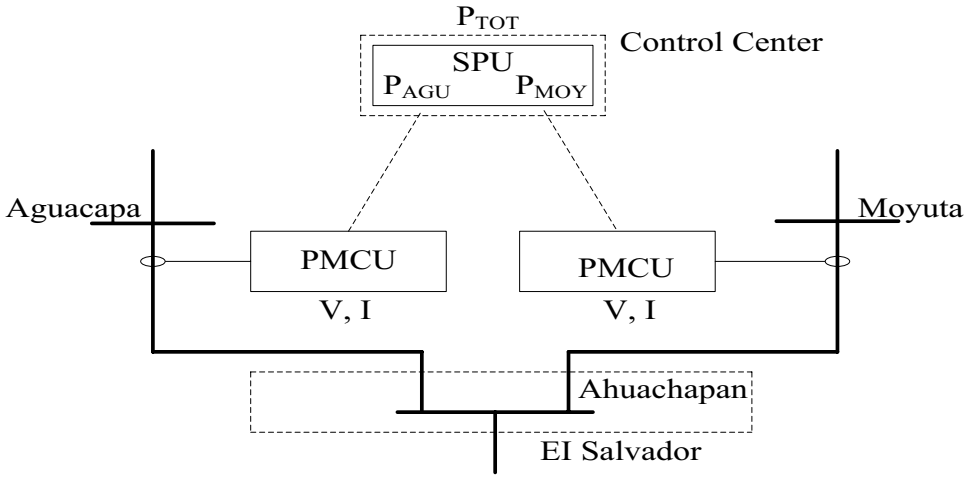


Fig. 15 PMCU network among Guatemala and El Salvador [152]

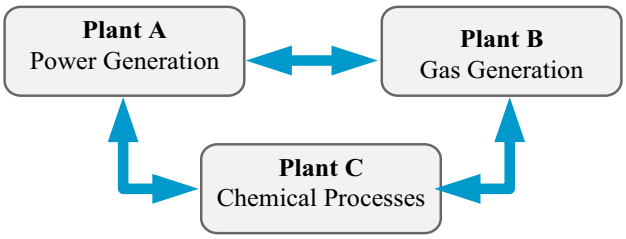


Fig. 16 Relationship between the plants in the complex [154]

operational history, the SCS identifies specific operational modes prone to instability. In response to these identified unstable states, the scheme activates the disconnection of El Salvador. Such strategic separation effectively isolates the Guatemalan power system from the broader Central American network, as depicted in Fig. 15 [152].

d) Brownfield Oil Production Facility

From a process perspective, managing this system within an existing oil production plant is essential to minimize operational disruptions while ensuring the system's reliability, stability, and expandability. The field employs a contemporary power management system (PMS) that effectively blends protection, control, automation, and communication tools. To bridge the gap between power demand and supply, the system features three unique backup load-shedding mechanisms, providing dynamic control critical for uninterrupted, well-coordinated operations. This capability enables the system to smoothly transition and adapt to diverse operational scenarios, ensuring a consistent and reliable energy supply. Additionally, the integration of a synchrophasor system enhances data resolution, enables real-time monitoring of power system dynamics, and facilitates post-event data archiving. The implementation and integration of the PMS are vital for preventing production disruptions at the brownfield facility and for bolstering the oil field's power system dependability.

e) Chemical Industrial Complex in *Mexico*

As depicted in Fig. 16, the chemical industrial complex consists of three interconnected plants, each reliant on at least one other for functionality. Failures in any plant can affect the entire complex. The complex has implemented a Contingencies-based Load-shedding Scheme, with the following objectives:

1. Ensure the secure and prioritized shutdown of processes involving chlorine and other hazardous gases, to prevent potentially fatal health risks near the plants during significant power outages.
2. Avoid unnecessary operational interruptions by selectively shedding non-essential loads in Plant B and Plant C.
3. Execute load shedding for specific loads in under 100 ms for 20 contingencies, minimizing power system disturbance impacts, maintaining system stability, and preventing cascading outages and complex blackouts by managing load demand before the activation of transformer and generator protection.

The complex operates efficiently, balancing generation and load requirements, with its connection to the local utility primarily for backup. The load-shedding strategy utilizes high-speed IEC 61850 Generic Object-Oriented Substation Event (GOOSE) messages and synchrophasors, facilitating equipment monitoring, remote load shedding, and precise power surveillance for on-site generators and their complex interconnections with the electric utility system. Since its introduction in April 2016, this contingency-based load-shedding strategy has proven reliable, particularly in

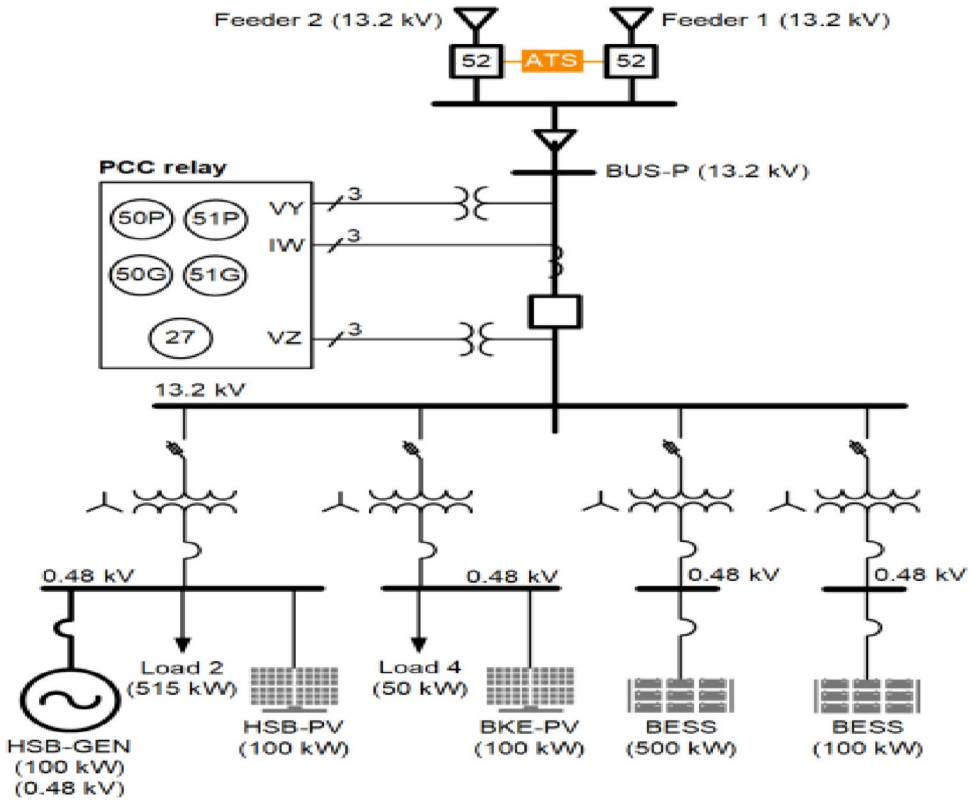


Fig. 17 Avista's protection scheme [155]

responding to faults in the 115 kV transmission line, with load-shedding times between 50 and 70 ms. The frequent power system faults and the consistent effectiveness of the load-shedding scheme have provided valuable insights for plant operators. This scheme's implementation has greatly improved the power system's reliability in the industrial complex, reducing process line interruptions and increasing production rates. Moreover, this enhanced reliability guarantees continuous power supply, creating a safer operational environment. Importantly, this reliable power supply plays a crucial role in ensuring safe shutdowns of processes involving chlorine and other hazardous gases [154].

f) Avista's Shared Energy Economy Project

The Avista Corporation has implemented modern Protection Automation and Control Schemes, incorporating energy storage management and a load-generation strategy, on a university campus with two buildings powered by a 13.2 kV utility supply. Each building features a battery, a rooftop photovoltaic (PV) system, and energy management with optimization capabilities. The Microgrid Control System (MCS) acts as a comprehensive layer, safeguarding, overseeing, and monitoring all assets within the microgrid's scope. The MCS supports various modes for microgrid operation, including rapid detection and isolation during islanding, optimization when connected to the grid, islanded mode operation, and strategies for distributed energy resource (DER) and load management. Furthermore, the MCS automatically synchronizes with the grid. As illustrated in Fig. 17, the microgrid power system can operate in tandem with the utility in grid-connected mode or autonomously in islanded mode. It maintains power system stability within defined limits, mitigating the impact of user errors or external control choices aimed at optimization, while protecting the integrity of the DER assets. The MCS underwent initial testing to confirm seamless integration before full-scale system tests. After successful validation, the Avista MCS was officially commissioned and is now operational [155].

Power grid automation systems are widely implemented across the globe, serving diverse purposes within the scope of their benefits. For instance, Gulf Petrochemical Industries have implemented these systems to achieve efficient isolation of critical infrastructure during disturbances. Southern California Edison's remedial Action Schemes (RAS) ensure the restoration of a reliable power supply following disruptions in their power transmission network. Additionally, the Phasor Measurement and Control Units (PMCUs) in Guatemala's power system are designed to disconnect or isolate the country's power system from the broader network during external disturbances. Both the Brownfield oil production company and the chemical industrial complex in Mexico have effectively utilized these systems to avert power disruptions to their critical plants, thereby mitigating potential

severe impacts on their entire operations and the safety of the surrounding communities.

Regulatory and Policy Considerations

Resilience in power systems is vital, as the economy of many countries depends on it [1]. A lack of resilience can lead to significant financial impacts on both nations and consumers [8], and in some instances, power system failures can even prove lethal, as seen in the case of the petrochemical industrial complex [154]. Therefore, the establishment of regulatory practices and policies supporting resilience in power systems is essential. The Federal Energy Regulatory Commission (FERC) advocates for policy advancement, emphasizing environmental considerations. These policies are crafted to promote distributed electricity infrastructure development, renewable energy generation expansion, and comprehensive climate change mitigation measures. Through these policies, FERC aims to create a sustainable energy landscape, highlighting eco-friendly practices and renewable resources to address climate change impacts. This approach reflects FERC's dedication to developing a resilient and eco-conscious energy sector prepared for future demands [156].

Reliability, a core principle of energy legislation, requires operators to fulfill extensive planning duties. These include managing power outages, ensuring power quality, and restoring frequency after deviations. Under EU law, Transmission System Operators (TSOs) must incorporate measures into their network development plans that guarantee system adequacy, security of supply, and support cross-border trade and consumption trend forecasting. Notably, the EU legislative framework regarding climate and energy financing has adopted resilience as a key factor in fund allocation, highlighting its proactive role in reinforcing power system resilience. This move demonstrates the EU's commitment to fortifying power infrastructures' robustness and adaptability, thereby promoting a power system that is not only reliable but also resilient in the face of various challenges. This commitment aligns with the overarching aim of a resilient, sustainable energy future [157]. National policies and federal preparedness initiatives underscore the importance of bolstering the flexibility of critical infrastructure, including electricity networks.

The 2013 issuance of Presidential Policy Directive 21 marked a notable shift in focus, expanding from protection against terrorism to resilience against various hazards like natural disasters, cyber incidents, and terrorism. This directive highlights the need for proactive, coordinated efforts to secure and enhance the resilience of critical infrastructure, including offering incentives to attract investors and increase power system resilience [23, 158]. According to "The Risk Management Process for Federal Facilities: An Interagency Security Committee Standard," it is essential to assess whether power grid automation systems within the business process network have comparable protection to the broader network. This assessment should include examining safeguards such as encryption, access management, and firewall implementation, for both the systems and the network infrastructure [159]. The European Union has mandated organizations to standardize the functionalities and communication interfaces of smart meters across various sectors. The Mandate M/441 aims to develop standards or technical documents, which include voluntary specifications and general rules for market-available products or systems. The main goals are to streamline smart metering system adoption, ensure interoperability, protect customers, and maintain system reliability [160]. Recognizing the commendable benefits of power system resilience, there is a pressing need for supporting policies and regulations.

Initiatives to foster resilience include providing incentives to mitigate investment risks in critical infrastructure and establishing regulations that offer investors a fair Return on Investment (ROI), aligning with consumer interests. The European Union's standards for electronic smart meters are examples of such measures to enhance system resilience. Organizations like the North American Electric Reliability Corporation (NERC) and the Federal Energy Regulatory Commission (FERC) are pivotal in setting reliability standards and developing systems to evaluate grid resilience, ensuring compliance with established requirements [161]. These regulatory policies also address cybersecurity risks, protecting critical infrastructure with standards and guidelines to secure power grid access from cyber-attacks, as seen in the NERC Critical Information Protection (CIP) mandates [162]. Unregulated integration of distributed energy resources (DERs) and renewables into power networks poses challenges, such as disruptions in dynamic load management, potentially impacting system reliability and resilience. Standards for interconnection and grid integration are vital for organized and adaptable integration of DERs [163]. Consumer-driven adjustments, facilitated by demand response and flexibility programs within a comprehensive regulatory framework, establish an efficient pricing mechanism. This encourages consumers to modulate their electricity usage, thereby enhancing the power system's resilience and reliability, especially during peak periods or emergencies [164]. In diverse economies with various power grid stakeholders, grid resilience is bolstered through collaborative efforts. Stakeholders share information and coordinate actions,

adopting shared best practices to strengthen the power system's resilience [165].

Regulatory Policy Models

Statistical analysis and modeling are crucial in understanding the impact and importance of regulatory policies on power grid resilience. These methods allow researchers and policymakers to assess the effects of regulatory interventions, gauge the effectiveness of policy measures, and identify opportunities for enhancing grid resilience [166]. As noted by [167], many studies examining the impact of weather conditions on power system resilience depend on the Markov analytical technique. However, this approach is better suited for small-scale networks. To address this limitation, the authors recommend a new simulation method designed for complex systems. This method aims to capture the stochastic behavior of weather patterns over time and space, using a sequential simulation Monte Carlo-based time series method with fragility curves. This approach provides a detailed analysis of the randomness and impact of weather-related events on various system components. However, it focuses exclusively on weather-related challenges. Contrastingly, [168] offers a thorough review of various models for power system resilience. This review includes models such as the Poisson distribution function, HAZUS flood models, precipitation rate, Yang Meng wind field model, and HAZUS earthquake model for weather-related challenges, as well as probabilistic failure models for component-level issues and regression and data mining models for system-level challenges.

Future Direction and Research Opportunities

Reference [169] introduced a two-layered scheme to bolster resilience against cyber-attacks. The first layer, rooted in blockchain expertise, manages the overall operation of the scheme. The second layer utilizes a reinforcement-learning approach to monitor system data and identify vulnerabilities. However, this scheme does not address uncertainties, such as new types of system disturbances, which should be considered in future enhancements.

The implementation of Big Data and IoT technologies in smart metering has enabled real-time sensing, processing, recording, and visualizing of electrical data within buildings. This system tracks the power consumption of individual appliances, but further development is required to enhance its predictive analytics capabilities. Such advancements would facilitate more effective analysis of consumer behavior, including occupancy detection based on electricity consumption patterns over time [170].

Reference [171] proposed a resilient control and monitoring method featuring a self-healing algorithm that sustains system operations amidst monitoring device or connection failures.

However, the existing context-aware monitoring (CAM) system falls short in analyzing and responding to feedback. Therefore, it is essential to upgrade the algorithm by integrating models and feedback controls that enable autonomous self-adaptation, reducing the need for manual intervention.

Developing an effective protection strategy for microgrids, especially in terms of its suitability for both grid-connected and islanded modes, poses significant challenges. Although extensive research has been conducted, existing studies highlight several gaps: a) Current strategies do not adequately protect diverse microgrids with varying configurations. b) Effective protection methods should distinguish high impedance faults, even when their current magnitudes resemble regular loads that don't cause significant voltage drops – an issue scarcely addressed in the literature. c) Most proposed methods cater to microgrids with radial feeders, leaving a gap in protecting microgrids with looped feeders [172].

Furthermore, [79] suggests enhancing the IoT architecture to better account for environmental conditions affecting remote management and monitoring. This entails integrating data collection and remote monitoring of the power transmission grid via IoT. Future research should focus on expanding interactions with other IoT systems, improving energy consumption optimization, developing a comprehensive big data management framework, refining machine learning algorithms for greater accuracy, enhancing system security measures, and ensuring the safety of structural elements.

Conclusion

Power systems confront major challenges from rare yet impactful natural disasters, such as hurricanes, tornadoes, and floods. These events significantly affect economies and individuals. To bolster resilience against such disruptions, strategies like targeted strengthening, defensive isolation, and network reconfiguration are utilized. Cyber threats also pose a substantial risk to power system integrity, exploiting the integration of sensors, monitoring tools, and communication devices. Deploying intrusion prevention systems, multi-layered defenses, and timely security protocols enhances the systems' resilience to these threats. Moreover, human errors and aging infrastructure components are notable causes of

power system failures, highlighting the need for vigilant maintenance and error prevention.

In response to the growing demand for clean energy and increased customer participation in energy management, the power sector is embracing modern technologies. Innovations such as IoT, Big Data, decentralized and distributed control systems (e.g., smart grids), AI/machine learning, 5G connectivity, and satellite technologies are being integrated to diversify and strengthen power system resilience. These technologies facilitate efficient sensing, extensive monitoring, smart metering, and sophisticated energy management.

The power system plays a pivotal role in any economy's development. While considerable efforts have been made to enhance its reliability and resilience, ongoing research and collaborative initiatives are essential. These efforts should focus on;

- Emerging threats,
- Technological advancements,
- Development of more effective policies, regulations, standards,
- Analysis and mitigation of dynamic risks.

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References

1. Bhusal N, Abdelmalak M, Kamruzzaman M, Benidris M (2020) Power system resilience: Current practices, challenges, and future directions. IEEE Access 8:18064–18086. https://doi.org/10.1109/ACCESS.2020.2968586
2. Dehghanian P, Aslan S, Dehghanian P (2017) Quantifying power system resiliency improvement using network reconfiguration. 2017 IEEE 60th International Midwest Symposium on Circuits and Systems (MWSCAS), Boston, MA, USA, pp. 1364-1367, https://doi.org/10.1109/MWSCAS.2017.8053185
3. Dagle J (2020) Resilience of Electric Power Infrastructure. Elsevier, pp. 69–77. https://doi.org/10.1016/B978-0-12-816240-8.00004-5
4. Mahzarnia M, Moghaddam MP, Baboli PT, Siano P (2020) A Review of the Measures to Enhance Power Systems Resilience. IEEE Syst J 14(3):4059–4070. https://doi.org/10.1109/JSYST.2020.2965993
5. Panteli M, Trakas DN, Mancarella P, Hatziargyriou ND (2017) Power Systems Resilience Assessment: Hardening and Smart Operational Enhancement Strategies. Proc IEEE 105(7):1202–1213. https://doi.org/10.1109/JPROC.2017.2691357
6. Phillips T, McJunkin T, Rieger C, Gardner J, Mehrpouyan H (2020) An Operational Resilience Metric for Modern Power Distribution Systems," 2020 IEEE 20th International Conference on Software Quality, Reliability and Security Companion (QRS-C), Macau, China, pp. 334–342, https://doi.org/10.1109/QRS-C51114.2020.00065.
7. Afzal S, Mokhlis H, Illias HA, Mansor NN, Shareef H (2021) State-of-the-art review on power system resilience and assessment techniques. IET Gener Transm Distrib 14(25):6107–6121. https://doi.org/10.1049/iet-gtd.2020.0531
8. Wang J, Gharavi H (2017) "Power Grid Resilience", Scanning the Issue. Proc IEEE 105(7):1199–1201
9. Beyza J, Yusta JM (2021) Integrated Risk Assessment for Robustness Evaluation and Resilience Optimisation of Power Systems after Cascading Failures. Energies 14(7):1–18. https://doi.org/10.3390/en14072028
10. Huang G, Wang J, Chen C, Qi J, Guo C (2017) Integration of Preventive and Emergency Responses for Power Grid Resilience Enhancement. IEEE Trans Power Syst 32(6):4451–4463. https://doi.org/10.1109/TPWRS.2017.2685640
11. Shittu E, Tibrewala A, Kalla S, Wang X (2021) Meta-analysis of the strategies for self-healing and resilience in power systems. Advances in Applied Energy 4:1–15. https://doi.org/10.1016/j.adapen.2021.100036
12. R. Ghaffarpour, M. J. Oskuee, S. Ravadanegh, and H. Alami, "Resiliency, an Answer to Existing Passive Defense Concerns About the Classical Reliability-Oriented View in Electric Power Networks," Passive Defense Quarterly, vol. 10, no. 1, pp. 1 – 22, 2019, magiran.com/p2005444
13. Lin Y, Bie Z, Qiu A (2018) A review of key strategies in realizing power system resilience. Glob Energy Interconnect 1(1):70–78. https://doi.org/10.14171/j.2096-5117.gei.2018.01.009
14. Bie Z, Lin Y, Li G, Li F (2017) Battling the Extreme: A Study on the Power System Resilience. Proc IEEE 105(7):1253–1266. https://doi.org/10.1109/JPROC.2017.2679040
15. K. Davis, "An Energy Management System Approach for Power System Cyber-Physical Resilience," *arXiv preprint*, pp. 1–2, 2021, [Online]. Available: https://arxiv.org/abs/2110.03451%0A, https://arxiv.org/pdf/2110.03451
16. Amani AM, Jalili M, Member S (2021) Power grids as complex networks : Resilience and reliability analysis. IEEE Access 9:119010–119031. https://doi.org/10.1109/ACCESS.2021.3107492
17. Arghandeh R, Meier AV, Mehrmanesh L, Mili L (2016) On the definition of cyber-physical resilience in power systems. Renew Sustain Energy Rev 58:1060–1069. https://doi.org/10.1016/j.rser.2015.12.193

18. Plotnek JJ, Slay J (2021) International Journal of Critical Infrastructure Protection Power systems resilience : Definition and taxonomy with a view towards metrics. Int J Crit Infrastruct Prot 33:100411. https://doi.org/10.1016/j.ijcip.2021.100411
19. Ratnam EL, Baldwin KGH, Mancarella P, Howden M, Seebeck L (2020) Electricity system resilience in a world of increased climate change and cybersecurity risk. Electr J 33(9):106833. https://doi.org/10.1016/j.tej.2020.106833
20. Guikema SD, Quiring SM, Han SR (2010) Prestorm Estimation of Hurricane Damage to Electric Power Distribution Systems. Risk Anal 30(12):1744–1752. https://doi.org/10.1111/j.1539-6924.2010.01510.x
21. Wang Y, Chen C, Wang J, Baldick R (2016) Research on Resilience of Power Systems Under Natural Disasters—A Review. IEEE Trans Power Syst 31(2):1604–1613. https://doi.org/10.1109/TPWRS.2015.2429656
22. Schweikert AE, Deinert MR (2021) Vulnerability and resilience of power systems infrastructure to natural hazards and climate change. WIREs Clim Change 24(12):1–14. https://doi.org/10.1002/wcc.724
23. Waseem M, Manshadi SD (2020) Electricity grid resilience amid various natural disasters: Challenges and solutions. Electricity Journal 33(10):106864. https://doi.org/10.1016/j.tej.2020.106864
24. Chen C, Wang J, Qiu F, Zhao D (2016) Resilient Distribution System by Microgrids Formation After Natural Disasters. IEEE Transactions on Smart Grid 7(2):958–966. https://doi.org/10.1109/TSG.2015.2429653
25. Chen C, Wang J, Ton D (2017) Modernizing Distribution System Restoration to Achieve Grid Resiliency Against Extreme Weather Events: An Integrated Solution. Proc IEEE 105(7):1267–1288. https://doi.org/10.1109/JPROC.2017.2684780
26. Li G, Huang G, Bie Z, Lin Y (2019) Component importance assessment of power systems for improving resilience under wind storms. J Mod Power Syst Clean Energy 7:676–687. https://doi.org/10.1007/s40565-019-0563-0
27. Henry D, Ramirez-marquez JE (2016) On the Impacts of Power Outages during Hurricane Sandy — A Resilience-Based Analysis. Syst Eng 19(1):59–75. https://doi.org/10.1002/sys.21338
28. Kwasinski A, Andrade F, Castro-Sitiriche MJ, O'Neill-Carrillo E (2019) Hurricane Maria Effects on Puerto Rico Electric Power Infrastructure. IEEE Power Energy Technol Syst J 6(1):85–94. https://doi.org/10.1109/JPETS.2019.2900293
29. A. Kocatepe, M. B. Ulak, L. M Konila, D Pinzan, E. E. Ozguven and R. Arghandeh. Co-resilience Assessment of Hurricane-induced Power Grid and Roadway Network Disruptions: A Case Study in Florida with a Focus on Critical Facilities" 2018 21st International Conference on Intelligent Transportation Systems (ITSC). https://doi.org/10.1109/ITSC.2018.8569574
30. N. Malla, S. Poudel, N. R. Karki and N. Gyawali, "Resilience of electrical power delivery system in response to natural disasters," 2017 7th International Conference on Power Systems (ICPS), Pune, India, 2017, pp. 806–811, https://doi.org/10.1109/ICPES.2017.8387400.
31. Ouyang M, Dueñas-osorio L (2014) Multi-dimensional hurricane resilience assessment of electric power systems. Struct Saf 48:15–24. https://doi.org/10.1016/j.strusafe.2014.01.001
32. Panteli M, Mancarella P (2015) Influence of extreme weather and climate change on the resilience of power systems: Impacts and possible mitigation strategies. Electr Power Syst 127:259–270. https://doi.org/10.1016/J.EPSR.2015.06.012
33. Panteli M, Pickering C, Wilkinson S, Dawson R, Mancarella P (2017) Power System Resilience to Extreme Weather: Fragility Modeling, Probabilistic Impact Assessment, and Adaptation Measures. IEEE Trans Power Syst 32(5):3747–3757. https://doi.org/10.1109/TPWRS.2016.2641463
34. Alhelou HH, Esmail M, Golshan H, Njenda TC (2019) A Survey on Power System Blackout and Cascading Events: Research Motivations and Challenges. Energies 12(4):682. https://doi.org/10.3390/en12040682
35. Espinoza S, Panteli M, Mancarella P, Rudnick H (2016) Multi-phase assessment and adaptation of power systems resilience to natural hazards. Electr Power Syst Res 136:352–361. https://doi.org/10.1016/J.EPSR.2016.03.019
36. Salman AM, Li Y, Stewart MG (2015) Evaluating system reliability and targeted hardening strategies of power distribution systems subjected to hurricanes. ReliabEngSystSaf 144:319–333. https://doi.org/10.1016/j.ress.2015.07.028
37. Panteli M, Trakas DN, Mancarella P, Hatziargyriou ND (2016) Boosting the Power Grid Resilience to Extreme Weather Events Using Defensive Islanding. IEEE Trans Smart Grid 7(6):2913–2922. https://doi.org/10.1109/TSG.2016.2535228
38 Dehghanian P, Aslan S, Dehghanian P (2018) Maintaining Electric System Safety Through An Enhanced Network Resilience. IEEE Trans Ind Appl 54(5):4927–4937. https://doi.org/10.1109/TIA.2018.2828389
39. Dehghanian P, Aslan S, "Enhancing electric safety by improving system resiliency in face of extreme emergencies," 2017 IEEE IAS Electrical Safety Workshop (ESW), Reno, NV, USA, 2017, pp. 1-1, https://doi.org/10.1109/ESW.2017.7914867
40. De Caro F, Carlini EM, Villacci D (2019) Flexibility sources for enhancing the resilience of a power grid in presence of severe weather conditions," 2019 AEIT International Annual Conference (AEIT), Florence, Italy, pp. 1–6, https://doi.org/10.23919/AEIT.2019.8893389.
41. Farzin H, Fotuhi-Firuzabad M, Moeini-Aghtaie M (2016) Enhancing Power System Resilience Through Hierarchical Outage Management in Multi-Microgrids. IEEE Trans Smart Grid 7(6):2869–2879. https://doi.org/10.1109/TSG.2016.2558628
42. Hussain A, Member S, Bui V, Member S (2018) A Proactive and Survivability-Constrained Operation Strategy for Enhancing Resilience of Microgrids Using Energy Storage System. IEEE Access 6:75495–75507. https://doi.org/10.1109/ACCESS.2018.2883418
43. Nazemi M, Moeini-Aghtaie M, Fotuhi-Firuzabad M, Dehghanian P (2020) Energy Storage Planning for Enhanced Resilience of Power Distribution Networks Against Earthquakes. IEEE Trans Sustain Energy 11(2):795–806. https://doi.org/10.1109/TSTE.2019.2907613
44. Kuntze N, Rudolph C, Leivesley S, Manz D, Endicott-Popovsky B (2014) Resilient core networks for energy distribution. 2014 IEEE PES General Meeting | Conference & Exposition, National Harbor, MD, USA, pp. 1-5, https://doi.org/10.1109/PESGM.2014.6939164
45. Nguyen T, Wang S, Alhazmi M, Nazemi M, Estebsari A, Dehghanian P (2020) Electric Power Grid Resilience to Cyber Adversaries: State of the Art. IEEE Access 8:87592–87608. https://doi.org/10.1109/ACCESS.2020.2993233
46. Amini S, Pasqualetti F, Mohsenian-Rad H (2018) Dynamic Load Altering Attacks Against Power System Stability: Attack Models and Protection Schemes. IEEE Trans Smart Grid 9(4):2862–2872. https://doi.org/10.1109/TSG.2016.2622686
47. Singh VK (2020) Design of Attack-Resilient System for Wide-Area Monitoring , Protection , and Control in Smart Grid Design of Attack-Resilient System for Wide-Area Monitoring , Protection , and Control in Smart Grid. IEEE Smart Grid, pp. 1–5
48. Wei L, Moghadasi AH, Sundararajan A, Sarwat AI (2015) Defending mechanisms for protecting power systems against intelligent attacks. 2015 10th System of Systems Engineering Conference (SoSE), San Antonio, TX, USA, pp. 12-17, https://doi.org/10.1109/SYSOSE.2015.7151941
49. McDonald J, Decroix H, Caire R, Sanchez J, Chollet S, Oualha N, Puccetti A (2013) Securuty Analysis and Risk Assessment Applied to the Electrical Distribution Network. 22nd International Conference on Electricity Distribution. Stockholm, 10–13
50. Mueller F, Bhattacharya S, Zimmer C. Cyber security for power grids, NCSU Secure Open Systems Initiative, [Online]. Available: http://moss.csc.ncsu.edu/~mueller/ftp/pub/mueller/papers/cpssec09.pdf

51. Stefanini A, Morch AZ, Dondosolla G, Deconinck G, Friessem P (2008) Current efforts concerning ICT security of the power grid. JRC Scientific and Technical Report, European Communities, Italy
52. Chu Z, Pinceti A, Kaviani R, Khodadadeh R, Li X, Zhang J, Saikumar K, Sahraei-Ardakani M, Mosier C, Podmore R, Hedman K, Kosut O, Sankar L (2021) A verifiable framework for cyber-physical attacks and countermeasures in a resilient electric power grid," arXiv:2104.13908v1 pp 1–9. Available: http://arxiv.org/abs/2104.13908
53. Technical. Report, "Cyber security and resilience guidelines for the smart energy operational environment". SyC-SE Cyber Security and Resilience Guidelines for Smart Energy final v2.docx. 1 ics.sans.org/media/E-ISAC_SANS_Ukraine_DUC_5.pdf
54. Grigoraş G, Bărbulescu C (2013) Human errors monitoring in electrical transmission networks based on a partitioning algorithm. Int J Electr Power Energy Syst 49(1):128–136. https://doi.org/10.1016/j.ijepes.2012.12.016
55. Liu Y, Luan Y, Zhang G, Hu H, Jiang J, Zhang L, Xi L (2020) Human reliability analysis for operators in the digital main control rooms of nuclear power plants. J Nucl Sci Technol 57(6):719–733. https://doi.org/10.1080/00223131.2020.1720848
56. Abreu JM, Hardy N, Cse F, Pereira FC, Zeifman M, Cse F (2015) Modeling Human Reliability in the Power Grid Environment : an Application of the SPAR-H Methodology. Sage J 59(1):662–666
57. Anokhin A, Alontseva E (2019) Implementation of Human – Machine Interface Design Principles to Prevent Errors Committed by NPP Operator," Proceedings of the 20th Congress of the International Ergonomics Association (IEA 2018). IEA 2018. Advances in Intelligent Systems and Computing, vol 822. Springer, Cham, pp. 743–753. https://doi.org/10.1007/978-3-319-96077-7
58. Bao Y, Guo J, Tang J, Li Z, Pang S, Guo C (2014) Analysis of power system operation reliability incorporating human errors. In: 17th International Conference on Electrical Machines and Systems (ICEMS), Hangzhou, China, pp 1052–1056. https://doi.org/10.1109/ICEMS.2014.7013625
59. Bao Y, Tang J, Wang Y, Guo C (2014) Quantification of human error probability in power system based on SLIM. 2014 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), Hong Kong, China, 1-5, https://doi.org/10.1109/APPEEC.2014.7066150
60. Bao Y, Guo C, Zhang J, Wu J, Pang S, Zhang Z (2018) Impact analysis of human factors on power system operation reliability. J Mod Power Syst Clean Energy 6(1):27–39. https://doi.org/10.1007/s40565-016-0231-6
61. Chul W, Ho T (2019) A classification of electrical component failures and their human error types in South Korean NPPs during last 10 years. Nucl Eng Technol 51(3):709–718. https://doi.org/10.1016/j.net.2018.12.011
62. Feng H, Li Y, Zhang D, Li J (2019) Research on Human Error Analysis in the Simulated Main Control Room of Nuclear Power Plant Based on EEG Brain Network. 2019 IEEE International Conference on Computational Intelligence and Virtual Environments for Measurement Systems and Applications (CIVEMSA), Tianjin, China, pp. 1–5, https://doi.org/10.1109/CIVEMSA45640.2019.9071568.
63. Wang Z, Rahnamay-Naeini M, Abreu JM, Shuvro RA, Das P, Mammoli AA, Ghani N, Hayat MM (2018) Impacts of Operators' Behavior on Reliability of Power Grids During Cascading Failures. IEEE Trans Power Syst 33(6):6013–6024. https://doi.org/10.1109/TPWRS.2018.2825348
64. Liu J, Guan Y, Qu X, Wang J (2023) Research on Human Factors Reliability of Electric Power Enterprises Based on HFCRA Model. 4th International Symposium on Frontiers of Economics and Management Science (FEMS 2023), vol. 169, pp. 1–5, 2023, https://doi.org/10.1051/shsconf/202316901001
65. Begovic MM, Peerzada A, Rohouma W, Balog R (2020) On accelerated aging of mechanical assets in distribution systems with renewable generation. In: Proceedings of the 53rd Hawaii International Conference on System Sciences (HICSS), pp 2961–2970. https://hdl.handle.net/10125/64104
66. Alvarez-Alvarado MS, Jayaweera D (2018) Aging Reliability Model for Generation Adequacy," 2018 IEEE International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), Boise, ID, USA, pp. 1–6. https://doi.org/10.1109/PMAPS.2018.8440529.
67. Awadalla SKE, Milanovic JV, Wang Z, Jarman PN (2015) Overview of power system reliability assessment considering age related failure of equipment. 2015 IEEE Power & Energy Society General Meeting, Denver, CO, USA, pp. 1-6, https://doi.org/10.1109/PESGM.2015.7285949.
68. Awadallah SKE, Milanović JV, Jarman PN (2014) Reliability Based Framework for Cost-Effective Replacement of Power Transmission Equipment. IEEE Trans Power Syst 29(5):2549–2557. https://doi.org/10.1109/TPWRS.2014.2309337
69. Ge H, Asgarpoor S, Hou J (2011) Aging equipment maintainability assessment for management of critical utility assets. 2011 IEEE Power and Energy Society General Meeting, Detroit, MI, USA, pp. 1-7, https://doi.org/10.1109/PES.2011.6039217
70. Nagarjun Y, Ganesh KL (2015) Field experiences with ageing electrical assets in power plants. 2015 International Conference on Condition Assessment Techniques in Electrical Systems (CATCON), Bangalore, India, pp. 7–12, https://doi.org/10.1109/CATCON.2015.7449499.
71. Pradhan P, Ahmad I, Habibi D, Kothapalli G, Masoum MAS (2020) Reducing the Impacts of Electric Vehicle Charging on Power Distribution Transformers. IEEE Access 8:210183–210193. https://doi.org/10.1109/ACCESS.2020.3040056
72. Man S, Yu M, Xie L, Yang M, He X (2021) Online impedance detection method for monitoring equipment aging in low voltage distribution system. International Conference on Sensors and Instruments (ICSI 2021), vol. 1188724, p. 62, 2021, https://doi.org/10.1117/12.2603015.
73 Xu R, Long D, Liu J, Yu W, Xu L (2021) Intelligent Assistant Decision-Making Method for Power Enterprise Customer Service Based on IoT Data Acquisition". Mob Inf Syst 2021:7100610. https://doi.org/10.1155/2021/7100610
74. Bagdadee AH, Hoque Z, Zhang L, Bagdadee AH, Hoque Z, Zhang L (2020) ScienceDirectScienceDirectIoT Based Wireless Sensor Network for Power Quality Control in IoT Based Wireless Sensor Smart Network for Power Quality Control in Grid Smart Grid. Procedia ComputSci 167(2019):1148–1160. https://doi.org/10.1016/j.procs.2020.03.417
75. Berouine A, Lachhab F, NaitMalek Y, Bakhouya M, Ouladsine R (2017) A Smart Metering Platform Using Big Data and IoT Technologies," 3rd International Conference of Cloud Computing Technologies and Applications (CloudTech), pp. 1–6. https://doi.org/10.1109/CloudTech.2017.8284729.
76. Hlaing W, Thepphaeng S, Nontaboot V, Tangsunantham N, Sangsuwan T, Pira C (2017) Implementation of WiFi-based single phase smart meter for Internet of Things (IoT). 2017 International Electrical Engineering Congress (iEECON), Pattaya, Thailand, pp. 1-4, https://doi.org/10.1109/IEECON.2017.8075793
77. Jadhav AR, Rajalakshmi P (2017) IoT enabled smart and secure power monitor. 2017 IEEE Region 10 Symposium (TENSYMP), Cochin, India, pp. 1-4, https://doi.org/10.1109/TENCONSpring.2017.8070096
78. LÓpez-Vargas A, Fuentes M, Vivar M (2018) On the application of IoT for real-time monitoring of small stand-alone PV systems: Results from a new smart datalogger. 2018 IEEE 7th World Conference on Photovoltaic Energy Conversion (WCPEC) (A Joint Conference of 45th IEEE PVSC, 28th PVSEC & 34th EU PVSEC), Waikoloa, HI, USA, pp. 605–607, https://doi.org/10.1109/PVSC.2018.8547612

79. Khoi NT, Ho T, Tien T (2020) An integrated data collection and remote monitoring of power transmission grid using Internet of Things (IoT). Proceedings of the 4th International Conference on Future Networks and Distributed Systems, St.Petersburg Russian Federation , pp.1–5, https://doi.org/10.1145/3440749.3442658
80. Pjevalica V, Pjevalica N, Petrović N (2019) Cloud Based Data Acquisition via IoT for Electric Power Quality Monitoring. 2019 Zooming Innovation in Consumer Technologies Conference (ZINC), Novi Sad, Serbia, pp. 82-85, https://doi.org/10.1109/ZINC.2019.8769386
81. Prabowo AS, Sudjoko RI (2020) The electrical energy usage of monitoring system at real-time using IoT as the primary policy of energy efficiency," International Conference on Advanced Mechanical and Industrial Engineering. https://doi.org/10.1088/1757-899X/909/1/012009
82. Cornel – Cristian A, Gabriel T, Călin-Arhip M, Zamfirescu A (2019) Smart grid integration of IoT. 2019 54th International Universities Power Engineering Conference (UPEC), Bucharest, Romania, pp. 1-5, https://doi.org/10.1109/UPEC.2019.8893551
83. Lee KT, Mun HK (2019) Real-time power monitoring using field-programmable gate array with IoT technology. IET Sci Meas Technol 13(6):931–935. https://doi.org/10.1049/iet-smt.2018.5692
84. Jain A, Jat DS (2020) An edge computing paradigm for time-sensitive applications. Fourth world conference on smart trends in systems, security and sustainability (WS4). London, UK, pp 798–803. https://doi.org/10.1109/WorldS450073.2020.9210325
85. Talbi K, El Ougli A, Tidhaf B, Zrouri H (2023) Low-cost real-time internet of things-based monitoring system for power grid transformers. Int J Electr Comput Eng 13(3):2579–2588. https://doi.org/10.11591/ijece.v13i3.pp2579-2588
86. Sai Shibu NB, Hanumanthiah A, Sai Rohith S, Yaswanth C, Hemanth Krishna P, Pavan JVS (2018) Development of IoT Enabled Smart Energy Meter with Remote Load Management. 2018 IEEE International Conference on Computational Intelligence and Computing Research (ICCIC), Madurai, India, pp. 1–4, https://doi.org/10.1109/ICCIC.2018.8782381
87. Sanya W, Bajpai G, Kombo O (2022) Twahirwa E (2022) Real-Time Data Analytics for Monitoring Electricity Consumption Using IoT Technology. Tanzania Journal of Engineering and Technology 41(1):28–37. https://doi.org/10.52339/tjet.vi.770
88. Hashim SM, Al-Mashhadani IB (2023) Adaptation of powerline communications-based smart metering deployments with IoT cloud platform. Indones J Electr Eng Comput Sci 29(2):825–837. https://doi.org/10.11591/ijeecs.v29.i2.pp825-837
89. Alhebshi F, Alnabilsi H, Alzebaidi J, Bensenouci A, Brahimi T, Bensenouci MA (2018)Monitoring the operation of transmission line in a smart grid system through IoT. 2018 15th Learning and Technology Conference (L&T), Jeddah, Saudi Arabia, pp. 139-146, https://doi.org/10.1109/LT.2018.8368498
90. Chen X, Sun L, Zhu H, Zhen Y, Chen H (2012) Application of Internet of Things in Power-Line Monitoring. 2012 International Conference on Cyber-Enabled Distributed Computing and Knowledge Discovery, Sanya, China, pp. 423–426, https://doi.org/10.1109/CyberC.2012.77
91. Hafidz I, Priyadi A, Pujiantara M, Anggriawan DO, Purnomo MH (2023) Development of IoT-Based Portable Power Quality Monitoring on Microgrids by Enhancing Protection Features. IEEE Access 11:49481–49492. https://doi.org/10.1109/ACCESS.2023.3276241
92. Koziel S, Hilber P, Ichise R (2019) Application of big data analytics to support power networks and their transition towards smart grids. 2019 IEEE International Conference on Big Data (Big Data), Los Angeles, CA, USA, pp. 6104–6106, https://doi.org/10.1109/BigData47090.2019.9005479
93. Bessa RJ (2018) Future Trends for Big Data Application in Power Systems. Big Data Application in Power Systems, pp. 223–242. https://doi.org/10.1016/B978-0-12-811968-6.00010-3
94. Angadi RV, Venkataramu PS, Daram SB (2020) Role of big data analytics in power system application, E3S web of conferences (ICMED) 184:1–7. https://doi.org/10.1051/e3sconf/202018401017
95. Zayas B, Espinosa A, Sanchez V, Perez J (2019) Getting ready for data analytics of electric power distribution systems. [Online]. Available: http://www.ineel.mx
96. Moradi, Shahinzadeh H, Nafisi H, Marzband M, Gharehpetian GB (2019) Attributes of big data analytics for data-driven decision making in cyber-physical power systems. In: 14th International Conference on Protection and Automation of Power Systems (IPAPS). Tehran, Iran, pp 83–92. https://doi.org/10.1109/IPAPS49326.2019.9069391
97. Kamińska-Chuchmała A (2019) Big Data application in power systems–brief overview of issues and proposals regarding the use of these data for prediction," RynekEnergii, [Online]. Available: https://www.cire.pl/pliki/2/2019/14___kaminska_chuchmala___zet19.pdf
98. Moharm K (2019) Advanced Engineering Informatics State of the art in big data applications in microgrid : A review. Adv Eng Inform 42:100945. https://doi.org/10.1016/j.aei.2019.100945
99. Nasir M, Jin Z, Khan HA, Zaffar NA, Vasquez JC, Guerrero JM (Feb.2019) A Decentralized Control Architecture Applied to DC Nanogrid Clusters for Rural Electrification in Developing Regions. IEEE Trans Power Electron 34(2):1773–1785. https://doi.org/10.1109/TPEL.2018.2828538
100. Li Z, Shahidehpour M, Aminifar F, Alabdulwahab A, Al-Turki Y (2017) Networked Microgrids for Enhancing the Power System Resilience. Proc IEEE 105(7):1289–1310. https://doi.org/10.1109/JPROC.2017.2685558
101. Kamps K, Möhrke F, Zdrallek M, Awater P, Schwan M (2020) Reliability of Decentralized Network Automation Systems and Impacts on Distribution Network Reliability. 2020 International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), Liege, Belgium, pp. 1–6, https://doi.org/10.1109/PMAPS47429.2020.9183449
102. Bajwa A, Mokhlis H, Mekhilef S, Mubin M (2019) Enhancing power system resilience leveraging microgrids : A review Enhancing power system resilience leveraging microgrids : A review. J Renew Sustain Energy, 1 – 11, https://doi.org/10.1063/1.5066264
103. Dagdougui H, Sacile R (2014) Decentralized Control of the Power Flows in a Network of Smart Microgrids Modeled as a Team of Cooperative Agents. IEEE Trans Control Syst Technol 22(2):510–519. https://doi.org/10.1109/TCST.2013.2261071
104. Colson CM, Nehrir MH, Gunderson RW (2011) Distributed multi-agent microgrids: a decentralized approach to resilient power system self-healing. 2011 4th International Symposium on Resilient Control Systems, Boise, ID, USA, pp. 83-88, https://doi.org/10.1109/ISRCS.2011.6016094
105. Ramchurn SD, Vytelingum P, Rogers A, Jennings N (2011) Agent-Based Control for Decentralised Demand Side Management in the Smart Grid," 10th International Conference on Autonomous Agents and Multiagent Systems, vol. 1, pp. 5–12
106. Nazemi M, Dehghanian P, Yang Z (2020) Swift Disaster Recovery for Resilient Power Grids: Integration of DERs with Mobile Power Sources," *2020 International Conference on Probabilistic Methods Applied to Power Systems, PMAPS 2020 - Proceedings*, pp. 1–6, https://doi.org/10.1109/PMAPS47429.2020.9183451
107. Dähling S, Kolen S, Monti A (2018) Swarm-based automation of electrical power distribution and transmission system support. IET Cyber-Physical Systems: Theory & Applications 3(4):212–223. https://doi.org/10.1049/iet-cps.2018.5001
108. Mahmood H, Jiang J (2019) Decentralized Power Management of Multiple PV, Battery, and Droop Units in an Islanded Microgrid. IEEE Transactions on Smart Grid 10(2):1898–1906. https://doi.org/10.1109/TSG.2017.2781468

109. Mezquita Y, Gil-González AB, Martín Del Rey A, Prieto J, Corchado JM (2022) Towards a Blockchain-Based Peer-to-Peer Energy Marketplace. Energies (Basel), 15(9). https://doi.org/10.3390/en15093046.
110. Tebekaemi E, Wijesekera D (2018) A Communications Model for Decentralized Autonomous Control of the Power Grid. 2018 IEEE International Conference on Communications (ICC), Kansas City, MO, USA, pp. 1–6, https://doi.org/10.1109/ICC.2018.8422911
111. Y. Chollot et al (2013) Decentralized self healing solution tested in the framework of GreenLys smart grid project. 2013 IEEE Grenoble Conference, Grenoble, France, pp. 1-7, https://doi.org/10.1109/PTC.2013.6652289
112. Bagdadee AH, Zhang L (2020) ScienceDirect Renewable energy based self-healing scheme in smart grid. Energy Rep 6:166–172. https://doi.org/10.1016/j.egyr.2019.11.058
113. Campos I, Saraiva F (2021) A Reactive Multi-agent System for Self-healing in Smart Distribution Grids. 2021 IEEE Congress on Evolutionary Computation (CEC), Kraków, Poland, pp. 209-216, https://doi.org/10.1109/CEC45853.2021.9505004
114. Li W, Deng Y, Zhang M, Li J, Chen S, Zhang S (2021) Integrated Multistage Self-Healing in Smart Distribution Grids Using Decentralized Multiagent. IEEE Access 9:159081–159092. https://doi.org/10.1109/ACCESS.2021.3131214
115. Farhoumandi M, Zhou Q, Shahidehpour M (2021) A review of machine learning applications in IoT-integrated modern power systems. Electricity Journal 34(1):106879. https://doi.org/10.1016/j.tej.2020.106879
116. Yousuf H, Zainal AY, Alshurideh M, Salloum SA (2021) Artificial Intelligence Models in Power System Analysis," Springer International Publishing, pp. 231–242. https://doi.org/10.1007/978-3-030-51920-9
117. Ibrahim MS, Dong W, Yang Q (2020) Machine learning driven smart electric power systems: Current trends and new perspectives. Appl Energy 272:115237. https://doi.org/10.1016/j.apenergy.2020.115237
118. Darab C, Tarnovan R, Turcu A, Martineac C (2019) Artificial Intelligence Techniques for Fault Location and Detection in Distributed Generation Power Systems. Proceedings of 2019 8th International Conference on Modern Power Systems, MPS 2019, pp. 1–4. https://doi.org/10.1109/MPS.2019.8759662
119. You S, Li H, Liu S, Sun K, Zhao Y, Xia H, Dong J, Su Y, Wang W, Cui Y, Yin H, Tan J, Liu Y (2023) 8 Build Smart Grids on Artificial Intelligence – A Real-world Example," in Applications of Big Data and Artificial Intelligence in Smart Energy Systems Smart Energy System: Design and its State-of-The Art Technologies : Volume 1, River Publishers, 2023, pp.193–220
120. Karimipour H, Dehghantanha A, Parizi RM, Choo K-KR, Leung H (2019) A Deep and Scalable Unsupervised Machine Learning System for Cyber-Attack Detection in Large-Scale Smart Grids. IEEE Access 7:80778–80788. https://doi.org/10.1109/ACCESS.2019.2920326
121. Sozontov A, Ivanova M, Gibadullin A (2019) Implementation of artificial intelligence in the electric power industry. Energy Res. J 9:1–6. https://doi.org/10.1051/e3sconf/201911401009
122. Upadhyay D, Manero J, Zaman M, Sampalli S (2021) Gradient Boosting Feature Selection with Machine Learning Classifiers for Intrusion Detection on Power Grids. IEEE Trans Netw Serv Manage 18(1):1104–1116. https://doi.org/10.1109/TNSM.2020.3032618
123. Maharjan L, Ditsworth M, Niraula M, Narvaez CC, Fahimi B (2019) Machine learning based energy management system for grid disaster mitigation. IET Smart Grid 2(2):172–182. https://doi.org/10.1049/iet-stg.2018.0043
124. Xia Y et al (2019) Anomaly Detection and Processing in Artificial Intelligence for IT Operations of Power System. 2019 IEEE 8th International Conference on Advanced Power System Automation and Protection (APAP), Xi'an, China, pp. 1099–1104, https://doi.org/10.1109/APAP47170.2019.9225115
125. C. Xinlei, Q. Rongfu, C. Yanli and X. Xinglang, "Power Grid Auxiliary Control System Based on Big Data Application and Artificial Intelligence Decision," 2020 International Conference on Artificial Intelligence and Computer Engineering (ICAICE), Beijing, China, 2020, pp. 154–157, https://doi.org/10.1109/ICAICE51518.2020.00036.
126. N. Zhang, W. Zhang, and Y. Shang, "Research on Integrated Energy System of Power Grid Based on Artificial Intelligence Algorithm of Machine Learning," IOP Conference Series: Earth and Environmental Science , vol. 714, 2021, https://doi.org/10.1088/1755-1315/714/4/042035.
127. Wang D, Wang X, Zhang Y, Jin L (2019) Detection of power grid disturbances and cyber-attacks based on machine learning. Journal of Information Security and Applications 46:42–52. https://doi.org/10.1016/j.jisa.2019.02.008
128. B. Bhattacharya and A. Sinha, "Intelligent fault analysis in electrical power grids," Proceedings - International Conference on Tools with Artificial Intelligence, ICTAI, vol. 2017-Novem, pp. 985–990, 2018, https://doi.org/10.1109/ICTAI.2017.00151.
129. L. Huan, H. Lishuai, G. Di, M. Fanbo, and W. Dongdong, "Research and Application of 5G Technology in Power System," 4th International Workshop on Renewable Energy and Development (IWRED 2020), pp. 1–8, 2020, https://doi.org/10.1088/1755-1315/510/2/022050.
130. Cruz AJT, Aybar-Mejía M, Roque YD, Ramírez KC, Durán JG, Weeks DR, Mariano-Hernández D, Hernández-Callejo L (2023) Implications of 5G Technology in the Management of Power Microgrids : A Review of the Literature. Energies 16(4):1–18. https://doi.org/10.3390/en16042020
131. Liu X, Chen B, Chen C, Jin D (2020) Electric power grid resilience with interdependencies between power and communication networks – a review. IET Smart Grid 3(2):182–193. https://doi.org/10.1049/iet-stg.2019.0202
132. Wikstrom G et al (2019) "Wireless Protection of Power Grids over a 5G Network", 2019 IEEE PES GTD Grand International Conference and Exposition Asia. GTD Asia 2019:976–981. https://doi.org/10.1109/GTDAsia.2019.8715864
133. R. Borgaonkar and M. G. Jaatun, "5G as an enabler for secure IoT in the smart grid," 2019 1st International Conference on Societal Automation, SA 2019, pp. 1–7, 2019, https://doi.org/10.1109/SA47457.2019.8938064.
134. G. Bag, L. Thrybom, and P. Hovila, "Challenges and opportunities of 5G in power grids," 24th International Conference & Exhibition on Electricity Distribution (CIRED), pp. pp. 2145–2148, 2017, https://doi.org/10.1049/oap-cired.2017.0374.
135. Hassebo A, Mohamed AA, Dorsinville R, Ali MA, "5G-based Converged Electric Power Grid and ICT Infrastructure,", (2018) IEEE 5G World Forum (5GWF). Silicon Valley, CA, USA 2018:33–37. https://doi.org/10.1109/5GWF.2018.8517041
136. Reka S, Dragi˘cevi´c T, Siano P, Sahaya Prabaharan SR (2019) Future Generation 5G Wireless Networks for Smart Grid : A Comprehensive Review. Energies 12(11):1–17. https://doi.org/10.3390/en12112140
137. Pavlovski M, Gajduk A, Todorovski M, Kocarev L (2017) Improving Power Grid Stability with Communication Infrastructure. IEEE J EmergSel Top Circuits Syst 7(3):349–358. https://doi.org/10.1109/JETCAS.2017.2672679
138. Deng C, Xiao X, Fu Z, Liu G, Yang H, Liu J (2011) Terrestrial-satellite hybrid backbone communication network for smart power grid," in Energy Procedia, Elsevier Ltd, pp. 27–36. https://doi.org/10.1016/j.egypro.2011.10.006
139. Dubey A, Bose A, Liu M, Ochoa LN (2020) Paving the Way for Advanced Distribution Management Systems Applications: Making the Most of Models and Data. IEEE Power Energ Mag 18(1):63–75. https://doi.org/10.1109/MPE.2019.2949442
140. Devanand P et al (2020) Advanced Distribution Management System: Improving Distribution Efficiency through an

Integrated Approach. IEEE Power Energ Mag 18(1):55–62. https://doi.org/10.1109/MPE.2019.2945345
141. Amiri SS, Rahmani M, McDonald JD (2021) An Updated Review on Distribution Management Systems within a Smart Grid Structure," 2021 11th Smart Grid Conference, SGC 2021, https://doi.org/10.1109/SGC54087.2021.9664067
142. Ebrahim MA, Wadie F, Abd-Allah MA (2020) An Algorithm for Detection of Fault, Islanding, and Power Swings in DG-Equipped Radial Distribution Networks. IEEE Syst J 14(3):3893–3903. https://doi.org/10.1109/JSYST.2019.2944870
143. Hossan MS, Chowdhury B (2018) Data-Driven Fault Location Scheme for Advanced Distribution Management Systems. IEEE Trans Smart Grid 10(5):5386–5396. https://doi.org/10.1109/TSG.2018.2881195
144. Fellah K, Abbou R, Khiat M (2021) Energy management system for surveillance and performance analysis of a micro-grid based on discrete event systems. Int J Green Energy 18(11):1104–1116. https://doi.org/10.1080/15435075.2021.1891909
145. Zahraoui Y et al (2021) Energy management system in microgrids: A comprehensive review," Sustainability 13(19). https://doi.org/10.3390/su131910492
146. Graa A, Benhamida F (2020) A Review On Optimization Methods Applied To Energy Management System. Serb J Manag 15(2):371–382. https://doi.org/10.5937/sjm15-22519
147. Javaid N, Hafeez G, Iqbal S, Alrajeh N, Alabed MS, Guizani M (2018) Energy Efficient Integration of Renewable Energy Sources in the Smart Grid for Demand Side Management. IEEE Access 6:77077–77096. https://doi.org/10.1109/ACCESS.2018.2866461
148. Sasaki Y, Yorino N, Zoka Y, Wahyudi F (2019) Robust stochastic dynamic load dispatch against uncertainties. IEEE Trans Smart Grid 9(6):5535–5542. https://doi.org/10.1109/TSG.2017.2690360
149. Qiu H, Gu W, Xu Y, Zhao B (2020) Multi-Time-Scale Rolling Optimal Dispatch for AC/DC Hybrid Microgrids with Day-Ahead Distributionally Robust Scheduling. IEEE Trans Sustain Energy 10(4):1653–1663. https://doi.org/10.1109/TSTE.2018.2868548
150. Al-Mulla A, Garg K, Manson S, El-Hamaky A (2010) System Islanding Using a Modern Decoupling System," Gulf Petrochemical Industries Company, 12th Annual Western Power Delivery Automation Conference Spokane, Washington
151. Dolezilek S (2010) Case Study of Mission-Critical Smart Grid Remedial Action Schemes Via Ethernet" Schweitzer Engineering Laboratories, 37th Annual Western Protective Relay Conference Spokane, Washington
152. Espinoza JV, Guzmán A, Calero F, Mynam MV, Palma E, Guatemala A (2017) Wide-Area Protection and Control Systems: A Collection of Technical Papers Representing Modern Solutions," Power and Energy Automation Conference
153. Hao K, Khatib AR, Herbert N, Shah N (2016) Case study: Integrating the power management system of an existing oil production field," in 2016 Petroleum and Chemical Industry Technical Conference, PCIC 2016, Institute of Electrical and Electronics Engineers Inc., https://doi.org/10.1109/PCICON.2016.7589245
154. Hernandez E, Hughes B, de Jesús Durón-Mendoza J (2020) Case study: Adaptive load shedding in critical industrial facilities," 15th International Conference on Developments in Power System Protection (DPSP 2020), Liverpool, UK, pp. 1–6, https://doi.org/10.1049/cp.2020.0117
155. Gibson J, Diedesch M, Mccoy T, Shah N, Paul TG, Upreti A (2022) Case Study: Implementing a Microgrid Protection and Control System for Avista's Shared Energy Economy Project." 75th Annual Georgia Tech Protective Relaying Conference Atlanta, Georgia
156. Phillips S (2019) Federal Regulation for a 'Resilient' Electricity Grid. Ecol Law Quart 46(2):415–454
157. Banet C (2023) Regulation and standards for a resilient european energy system
158. Mukhopadhyay S, Hastak M (2016) Public Utility Commissions to Foster Resilience Investment in Power Grid Infrastructure. Procedia SocBehavSci 218:5–12. https://doi.org/10.1016/j.sbspro.2016.04.005
159. Presidential Policy Directive 21 Implementation: an interagency security committee white paper (2015), pp 1–15
160. Joint E, Group W (2011) Final report Standards for Smart Grids,.
161. Brown SW (2019) Leveraging Mandatory and Enforceable Electric Reliability Standards for Non-Registered Rural Electric Utilities. Available: http://www.nerc.com/files/10_Benchmarks_of_Excellent_Reliability_
162. Bose D, Chanda CK, Chakrabarti A (2020) Vulnerability assessment of a power transmission network employing complex network theory in a resilience framework. Microsyst Technol 26(8):2443–2451. https://doi.org/10.1007/s00542-020-04785-x
163. Wichita State University (2019) IEEE Power & Energy Society, and Institute of Electrical and Electronics Engineers, 51st North American Power Symposium : Wichita State University
164. Dadkhah A, Vahidi B, Shafie-Khah M, Catalão JPS (2020) Power system flexibility improvement with a focus on demand response and wind power variability. IET Renew Power Gener 14(6):1095–1103. https://doi.org/10.1049/iet-rpg.2019.0123
165. Lin Y, Bie Z, Qiu A (2019) A review of key strategies in realizing power system resilience". Global Energy Interconnection 1(1):70–78. https://doi.org/10.14171/j.2096-5117.gei.2018.01.009
166. Shen L, Cassottana B, Tang LC (2019) Statistical trend tests for resilience of power systems. ReliabEngSystSaf 177:138–147. https://doi.org/10.1016/j.ress.2018.05.006
167. Panteli M, Mancarella P (2017) Modeling and Evaluating the Resilience of Critical Electrical Power Infrastructure to Extreme Weather Events. IEEE Syst J 11(3):1733–1742. https://doi.org/10.1109/JSYST.2015.2389272
168. Oster M, Chatterjee S, Pan F, Bakker C, Bhattacharya A, Perkins C (2020) Power system resilience through defender-attacker-defender models with uncertainty: An overview," in 2020 Resilience Week, RWS 2020, Institute of Electrical and Electronics Engineers Inc., pp. 11–17. https://doi.org/10.1109/RWS50334.2020.9241279
169. Chen J, Mohamed MA, Dampage U, Rezaei M, Salmen SH, AI Obaid S, Annuk A (2021) A Multi-Layer Security Scheme for Mitigating Smart Grid Vulnerability against Faults and Cyber-Attacks. Appl Sci 11(21). https://doi.org/10.3390/app11219972
170. Berouine A, Lachhab F, Malek YN, Bakhouya M, Ouladsine R (2017) A smart metering platform using big data and IoT technologies. 2017 3rd International Conference of Cloud Computing Technologies and Applications (CloudTech), Rabat, Morocco, pp. 1-6, https://doi.org/10.1109/CloudTech.2017.8284729
171. Hauer D, Ratasich D, Krammer L, Jantsch A (2020) A Methodology for Resilient Control and Monitoring in Smart Grids," 2020 IEEE International Conference on Industrial Technology (ICIT), Buenos Aires, Argentina, pp. 589–594, https://doi.org/10.1109/ICIT45562.2020.9067283
172. Mirsaeidi S, Mat D, Wazir M, Hafiz M, Ghaffari K (2014) Electrical Power and Energy Systems An analytical literature review of the available techniques for the protection of micro-grids. Int J Electr Power Energy Syst 58:300–306. https://doi.org/10.1016/j.ijepes.2014.01.032

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